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Operations Research

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Conceptual Framework, Technical Evolution, and Future Scalability

Literature Review, Mathematical Model, and Limitations

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Executive Summary

1. Context and Nature of the Document

This document constitutes an early version and represents the original theoretical approach conceived at the beginning of the project. It is explicitly included for the purpose of ensuring transparency in the design process and documenting the first sketch that was intended to be formally established. Preserving and presenting this section is an academically defensible practice, as it illustrates the evolution of analytical thinking and the methodological trajectory of the team.

2. Scientific Continuity and Literature Review

It is important to emphasize that this document shares exactly the same methodological foundations as the final executed version. The Literature Review section described herein remains fully valid and serves as the indispensable structural axis to define the boundaries of the logistics problem. This conceptual framework was the foundation that helped us optimally organize all the information, parameters, and data that currently support the project and are consolidated in our GitHub repository.

3. Technical Complexity and Python Implementation

The mathematical formulation and the level of modeling reflected in this version represent a more complex structure, which required technical and programming skills superior to those acquired at the beginning of the practical coding phase. Therefore, during the development stage, a transition toward a simplified and short-term executable model was chosen.

Nevertheless, this document is presented as a comprehensive piece of great informational and methodological value: it clearly demonstrates the scope of what we are capable of structuring conceptually and defines the type of advanced solutions that, in the future, can be coded and implemented through the strengthening and acquisition of enhanced skills in Python code development.

4. Future Outlook and Scalability

Far from being a discarded design, this ambitious framework serves as our roadmap for subsequent phases. By establishing this robust theoretical foundation, the project maintains a clear scalability plan. Once technical competencies and the use of advanced solvers are consolidated, the formulations contained herein will serve as the direct architecture to evolve the current code into the high-fidelity optimization system that was originally projected.

I. Introduction

Lidl is one of the largest and most influential discount supermarket chains in Germany, operating an extensive retail network characterized by high-frequency replenishment cycles. Market reports indicate that Lidl consistently maintains a leading market share in the German discount retail sector, generating tens of billions of euros in annual revenue (*Discount Retail Consulting GmbH, 2025).* The vast geographic distribution of these retail outlets with high inventory turnover, introduces substantial logistical complexities for suppliers and distribution networks.

Concurrently, Dr. Oetker operates as one of Germany's premier food manufacturers, distributing a diverse product portfolio to supermarkets and retail hubs nationwide within this supply chain. Therefore, highly efficient transportation planning is crucial to guarantee delivery reliability, maintain on-shelf availability, and mitigate operational issues that can lead to increased costs and time delivery.

In regional distribution networks, particularly within the Ostwestfalen-Lippe (*OWL*) area, logistical planning is further constrained by strict operational requirements and *stochastic factors** (App.1), including urban traffic congestion and driver-related ergonomic considerations that must be accounted for to ensure efficient service delivery. In this context, heterogeneous vehicle fleets introduce additional complexities such as varying payload capacities, working-time regulations, and customer-specific delivery time windows. These characteristics reflect realistic operational conditions commonly encountered in practice, and they motivate the formulation of more complex mathematical models within the field of Operations Research (OR), as well as the development of efficient algorithmic implementations for their solution, consequently, resulting in a more precise interpretation and analysis to find the optimal solution to the problem.

The Ostwestfalen-Lippe (OWL) area* (App.2) is considered as a representative case study for the analysis of complex logistics operations. In this context, logistical planning is subject to a variety of operational constraints and stochastic factors, including urban traffic congestion and driver-related ergonomic considerations, all of which must be accounted for to ensure efficient and reliable service delivery. Moreover, heterogeneous vehicle fleets introduce additional complexity through varying payload capacities, working-time regulations, and customer-specific delivery time windows; otherwise, if these factors are not considered, it may lead to inefficient routing directly: <u>prolonged delivery times, inflated operational overhead, and elevated environmental emissions.</u>

These characteristics reflect realistic operational conditions and provide a suitable framework for formulating a richer and more constrained optimization model within the field of *OR*. The incorporation of such restrictions leads to a more comprehensive mathematical formulation and supports the development of more advanced solution approaches, allowing the problem to be addressed in a way that is closer to real-world decision-making scenarios. At the same time, this formulation serves as an application of the concepts and methodologies acquired during the course, enabling their integration in a complex, realistic optimization setting.

Furthermore, logistics planning for this case study cannot be isolated of selecting the shortest path between a centralized depot and an individual store. Instead, it requires the design of coordinated multi-stop routes for a heterogeneous fleet while simultaneously considering multiple and often conflicting operational objectives, such as cost efficiency, service reliability, and resource utilization. As the number of customers and feasible routing combinations increases, the problem size grows combinatorially, significantly increasing computational complexity and making manual planning approaches unsuitable for by-hand problem solving; hence, this present document will also show the use and analysis by computational solvers.

To address these challenges, this project develops a sustainability-oriented logistics optimization analysis. By evaluating how mathematical routing decisions can reduce unnecessary travel distance and fuel consumption, the resulting optimization model serves as a robust decision-support system. It enables logistics planners to identify strategic delivery sequences and fleet configurations that maximize operational efficiency without compromising service quality.

Project Question

Which delivery sequence and vehicle capacity configuration (low-, medium-, or high-capacity vehicles) should be selected to distribute Dr. Oetker products to the 24 Lidl stores in the Ostwestfalen-Lippe region in order to minimize transportation time and operational costs while satisfying delivery time windows, vehicle capacities, and ergonomic constraints?

Main objective

The objective of the model is to **minimize total operational costs**, encompassing vehicle usage, distance travelled and fuel consumption, while maintaining distribution efficiency and meeting required delivery timelines.

II. Literature Review

2.1. Classification within the Operations Research Domain

This project addresses a combinatorial optimization problem within the field of *Rich Vehicle Routing Problems (RVRPs*), a class of models introduced in the *Operations Research (OR)* literature to represent routing problems that extend beyond the assumptions of the classical *Vehicle Routing Problem (VRP*); unlike traditional *VRP* (first introduced by Datzing and Ramser in 1959 in the “Traveling Salesman Problem), *RVRPs* incorporate multiple real-world constraints, such as strict delivery time windows, limited vehicle capacities, ergonomic considerations (for example driver-service ergonomic related constraints), multiple depots, customer priorities, and traffic or road restrictions (Cáceres Cruz, et al. 2014). By considering this practical focus, *RVRP* models enable the development of mathematical formulations and solution algorithms that more accurately reflect real distribution systems, producing solutions that are both operationally feasible and applicable in industrial settings.

Specifically, this project focuses on the *Heterogeneous Fleet Vehicle Routing Problem with Time Windows (HFVRPTW)*, a variant of the *RVRP* that considers fleets composed of vehicles with different capacities and operational characteristics while satisfying customer time-window constraints. The proposed mathematical model was developed following an extensive review of the relevant *OR* literature, with particular emphasis on studies addressing heterogeneous fleets and time-constrained routing problems described in the bibliography section.

In high-frequency resupply networks, such as the distribution system analyzed in this present document, ignoring differences in vehicle capacities and urban accessibility constraints may lead to substantial operational inefficiencies like poor vehicle utilization, increased transportation costs, unnecessary trips, violations of delivery time windows, and the assignment of vehicles that are unable to access certain delivery locations due to road regulations, low-emission zones, weight restrictions, or limited urban infrastructure. As a result, the simplifying assumptions of the classical *VRP* is scarce to represent the operational complexity of modern distribution systems, making richer formulations such as the *HFVRPTW* a more appropriate modeling framework.

Furthermore, with a mathematical approach, the *HFVRPTW* is part of a *NP-hard formulations* according to (Gendreau et al., 1999) since it amalgams two type of sub-combinational problems of exponential scale:

1. Bin Packing Problem (BPP): Decide with sub-set of clients and demands are assigned to which sub-set of clients for each demand without exceeding its volumetric or maximum weight capacity (Q_k).
2. Traveling Salesman Problem with Time Windows (TSPTW): Determine the optimal sequence of visit (arc-sequence) for each activated vehicle guaranteeing that the time arrival (t_{ik}) is in the correct time interval for each of the nodes of the system.

2.2 Main pillars of this HFVRPTW case study model

- Time Windows

The formal integration of temporal variables and restrictive service time windows in vehicle routing was consolidated by Solomon (1987), whose work *Algorithms for the Vehicle Routing and Scheduling Problems with Time Window Constraints* remains a key reference in *OR* literature for benchmarking routing algorithms. In his study, Solomon showed that introducing time windows significantly reduces the feasible solution space. However, even though this reduction might suggest a simpler problem, it actually introduces additional computational difficulty when solving it with exact or heuristic methods. In particular, determining whether a feasible solution exists becomes more complex due to the mismatch between spatial proximity and temporal compatibility: two customers that are geographically close may still be incompatible within the same route if their service time windows do not overlap.

- Heterogeneous Fleet and Cost Offsets

Some authors like Gendreau et al. (1999) and, subsequently, Baldacci et al. (2008) formalized the Heterogeneous Fixed Fleet VRP (*HFFVRP*) and the Fleet Size and Mix VRP (*FSMVRP*), respectively. In their work, they showed that the objective function of a realistic *OR* model cannot be restricted to exclusively minimizing total travel distance alone. Instead, it must capture a fundamental cost trade-off structure.

The authors demonstrated the existence of a mathematical trade-off between fixed and variable costs associated with different vehicle types. For instance, deploying a heavy articulated truck involves a high fixed operational cost (FixedCost□), but benefits from a lower variable cost per pallet-kilometer (VariableCost□). In contrast, a light rigid vehicle has a significantly lower activation cost, but reaches its capacity limits much earlier, which increases the number of required trips and, consequently, the total variable cost.

- Heuristic algorithms

As computational power increased and gained relevance in *OR* formulations, researchers began to integrate more complex and realistic constraints into routing models. A clear example is Cáceres-Cruz et al. (2014), who mapped the state of the system in *RVRPs*, concluding that the combination of *HFVRPTW* represents one of the most practically relevant frontiers in transportation decision support systems (DSS).

Although their investigations also showed that when these constraints are incorporated into exact mathematical solvers, computational times grow exponentially. This scalability limitation is particularly critical in real-world applications, where decision-making must be performed within short time frames. For this reason, heuristic and metaheuristic algorithms are essential, as they are designed to efficiently explore the solution space and generate high-quality near-optimal solutions in reasonable computational times, especially for instances exceeding 25 to 30 nodes, where exact methods often become computationally intractable.

- Food Logistics and Retail Context in Germany

In order to maintain consistency with the real operational conditions of Dr. Oetker and Lidl considered in this study, Holzapfel et al. (2016) analyzed the distribution structures of discount supermarket chains in Germany. Their study showed that the time windows imposed by retailers such as Lidl are typically “hard time windows” (strict delivery time intervals that must be satisfied, otherwise the solution is infeasible) due to the high level of synchronization required at unloading docks (cross-docking operations), as well as German regulations on driving and rest times for professional drivers (*Fahrpersonalverordnung – FPersV).*

Furthermore, they confirmed that fleet heterogeneity is not merely a modeling assumption but a physical requirement in practice, since certain urban centers restrict the access of trucks exceeding 12 tons during daytime hours. This restriction forces the use of smaller commercial vehicles for last-mile distribution or requires transshipment operations between heavy and light-duty vehicles.

2.3. Modeling Approaches in Rich Vehicle Routing

The mathematical formulation of the *Heterogeneous Fleet Vehicle Routing Problem with Time Windows (HFVRPTW)* can be approached through different methodological paradigms depending on how the decision domain, vehicle tracking, and constraints are structurally defined. According to the foundational literature in routing optimization (Toth & Vigo, 2014; Irnich et al., 2014), three primary mathematical modeling frameworks exist to map these complex distribution environments: *the Three-Index Formulation, the Two-Index Formulation, and the Set Partitioning Formulation*, which are explained in the following bullet-points:

- **Three-Index Formulation (Arc-Based Models):** This framework is widely recognized as the standard extension of the classical vehicle routing problem when dealing with non-homogeneous assets (Irnich et al., 2014). This model relies on a binary decision variable x_{ijk}, which equals 1 if vehicle k directly traverses the directed arc *(i,j)* connecting two physical nodes within the spatial network, and 0 otherwise. To couple spatial routing with temporal constraints, this approach incorporates continuous tracking variables, traditionally defined as t_{ik} representing the exact arrival time of vehicle k at node i.
    - This approach has the advantage that when introducing vehicle index k it forces the decision space to scale <u>quadratically with the number of customers and linearly with the total fleet size</u>, leading to an overall variable complexity. This implies a controlled complexity by maintaining a strictly polynomial growth of $O\ (N^2 \bullet K)$. Where "O" represents the upper bound of the model's size multiplying the spatial combinations (BIG-O), "N^2" the total number of arcs (origin-destination pairs) connecting the N logistics nodes in the networks, and since any node i can connect to any node j, the decision grows quadratically, while "k" represents the heterogeneous fleet.
    - This avoids the exponential or factorial explosion inherent in routing problems, meaning that the decision space is mathematically bounded. Consequently, instead of forcing the solver to naively explore an unmanageable *N!* combinatorial sequence, the problem is mapped onto a finite and structured set of binary variables.
    - **Matrix Relaxation Weakness:** Subtour elimination constraints and time window integrations are explicitly <u>modeled via extensive Big-M formulations.</u>

However, these linear continuous relaxations are notoriously loose. Consequently, standard commercial mixed-integer linear programming (MILP) solvers struggle to prune the branch-and-bound tree effectively, making this approach computationally impossible for instances scaling beyond 20 to 30 customer nodes. Despite this limitation, this approach remains highly favored in VRP literature when exact operational control over heterogeneous vehicle attributes on explicit arcs is non-negotiable.

- **Two-Index Formulation:** To mitigate the severe coordinate-space expansion of three-index formulations, modern adaptations frequently collapse individual vehicle tracking by aggregating the heterogeneous fleet into distinct, homogeneous vehicle classes or types (Toth and Vigo, 2014). The principal binary decision variable is reduced to x_{ijm}, which denotes whether any vehicle belonging to class m traverses the arc (i,j).Instead of relying on continuous time variables per vehicle, subtour elimination and capacity utilization are handled implicitly by transforming the routing architecture into a multi-commodity network flow problem. Here, continuous variables represent the flow of physical load units or accumulated elapsed time along the arcs. While this drastically compresses memory requirements and reduces the overall integer variable count to $O(N^2 \bullet M)$ (where the number of classes M is significantly smaller than the total fleet size K), it introduces complex multi-commodity tracking constraints. Furthermore, managing individual vehicle fixed activation costs ($FixedCost_k$) becomes a complex task if a specific class contains a finite, fixed number of assets rather than an infinite pool, often requiring additional cutting planes to prevent the over-allocation of specialized vehicles.
- **Set Partitioning Formulation (Route-Based Models):** This paradigm fundamentally changes the decision domain by shifting the optimization from arc selections to entire path trajectories.
    - Let Ω represent the complete set of all feasible routes that a heterogeneous vehicle type can execute while completely satisfying capacity limits and hard time windows. The model utilizes a binary decision variable y_r, which equals 1 if route r $\in$ Ω is selected in the final logistics plan, and 0 otherwise. This structure shifts the mathematical complexity from the solver's branch-and-bound tree to the route-generation phase, serving as the foundation for modern exact decomposition methods.
        - The set partitioning formulation provides a significantly tighter Linear Programming (LP) relaxation bound compared to arc-based models, which drastically accelerates the branch-and-bound search tree of professional solvers by providing precise lower bounds almost instantaneously. Furthermore, because entire valid path trajectories are calculated externally during the route-generation phase (Ω), this approach completely eliminates the need for numerically unstable Big-M equations, sub-tour elimination parameters, and complex scheduling constraints within the optimization matrix. This

mathematical decoupling allows researchers to easily integrate highly restrictive, non-linear real-world logistics parameters (such as time-dependent congestion multipliers $\mu(t)$, vehicle-specific urban accessibility rules, and strict labor laws) directly into the path generation algorithm, effectively isolating the master optimization model from operational complexity.

- The primary mathematical drawback of this paradigm is the curse of dimensionality, as the complete set of feasible routes scales factorially or exponentially with the number of logistics nodes, causing severe memory exhaustion and computational failure in standard commercial solvers during large-scale stress tests. Consequently, this model cannot be solved directly through traditional matrix loops and instead forces the implementation of advanced exact decomposition frameworks, such as Column Generation and custom Branch-and-Price algorithms, which significantly increases code complexity and development time. Additionally, the model exhibits a severe sensitivity to constraint relaxation (meaning that if store time windows or fleet capacities are expanded, the variable space explodes uncontrollably) while the continuous LP relaxation frequently yields highly fractional path distributions that require extensive branching routines to convert into practical, integer-feasible schedules.

While the Set Partitioning paradigm is widely recognized in state-of-the-art research as the mathematically superior approach due to its exceptionally tight linear programming (LP) relaxation bounds, examining all three frameworks is critical for operations research analysis. Evaluating these alternative formulations provides deep insights into the underlying solution space, demonstrating how a single commercial solver (such as Gurobi) alters its computational performance, variable dimensionality, and branching efficiency when a single logistics problem is mapped using different mathematical structures.

2.4 Typical Constraints in Real-World Applications

While the classical *Vehicle Routing Problem (VRP)* simplifies outbound logistics by assuming unrestricted urban access, homogeneous fleets, and flexible delivery timing, industrial distribution networks must integrate a multi-layered matrix of operational, physical, and legal restrictions to accurately replicate real-world supply chain dynamics. In operations research literature, these binding requirements are classified into several fundamental categories:

- Temporal Constraints and Service Dynamics: These primarily include Hard Time Windows, where early arrival forces individual vehicles to wait at the customer node, and late arrival causes immediate mathematical infeasibility (Bräysy & Gendreau, 2005). Conversely, Soft Time Windows permit marginal violations but penalize them financially within the objective function to evaluate corporate cost-service trade-offs (Figliozzi, 2010). Furthermore, realistic models must distinguish between pure transit and node operations by incorporating Service Times. In consumer goods distribution, this comprises a fixed administrative setup duration combined with a variable discharge time directly proportional to the volume delivered, ensuring that scheduling calculations account for the physical presence of the vehicle at the loading dock.

- Time-Dependent Volatility (Traffic Congestion): Traditional frameworks frequently assume static travel times along network arcs; however, contemporary logistics research recognizes this as a significant scheduling risk. In dense industrial regions, deterministic travel times introduce severe vulnerability to time-window violations and cold-chain disruption. Consequently, advanced applications incorporate Time-Dependent Vehicle Routing (TDVRP) paradigms, scaling base travel times by a non-linear or piecewise step-function congestion multiplier, dictated by the exact departure epoch from the origin vertex to accurately map peak commuting hours (Ichoua et al., 2003).
- Fleet Capacity Constraints: In a Heterogeneous Fleet VRP (HFVRP), each vehicle possesses a strict upper bound regarding its physical carrying capacity, measured in weight, volume, or discrete handling units such as standardized pallets. The mathematical formulation must enforce that the cumulative demand allocated to any individual route trajectory does not exceed the volumetric capacity of the assigned vehicle asset, driving the trade-off between dispatching fewer high-capacity multi-axle trucks versus multiple smaller units (Toth & Vigo, 2014).
- Physical and Infrastructure Compatibility: Vehicles face strict spatial accessibility constraints based on physical customer architecture and localized municipal regulations (Cáceres-Cruz et al., 2014). Urban centers frequently enforce restricted access parameters that penalize or completely prohibit large-scale heavy freight transport due to narrow historic roads, low-clearance bridges, or maximum weight thresholds (e.g., maximum 12 tons). Furthermore, strict environmental frameworks, such as urban Low Emission Zones (LEZ), impose multi-tier access rules based on vehicle fleet emission ratings (Holzapfel et al., 2016). Finally, specialized product lines (such as temperature-controlled food distribution) impose strict hardware compatibility, requiring refrigerated compartment assets to preserve product integrity.
- Driver and Labor Regulations: Real-world transport optimization must operate within rigid legislative frameworks concerning human capital. Within Germany and the broader European Union, commercial logistics must strictly comply with driving, break, and rest-period allocations governed by the Fahrpersonalverordnung (FPersV) and EC Regulation No 561/2006 (Holzapfel et al., 2016). These legal structures impose a hard upper bound on the maximum allowable driving shift and overall route duration, preventing driver fatigue and transforming standard routing matrices into highly constrained scheduling environments (Goel, 2009).

2.5 Exact Solver Approaches in the Literature

To solve the combinatorial complexity of the Heterogeneous Fleet Vehicle Routing Problem with Time Windows (*HFVRPTW*), exact optimization frameworks have evolved significantly.

While classical mathematical models, such as the three-index arc-based formulation can theoretically be solved directly by commercial Branch-and-Bound algorithms (e.g., Gurobi, CPLEX), they frequently experience severe "matrix relaxation weakness" due to loose Big-M bounds, like it was mentioned above. Consequently, modern ORliterature relies on specialized exact methodologies designed to leverage tighter Linear Programming (LP) relaxations or advanced decomposition techniques. The most prominent exact methods include:

- Branch-and-Cut (B&C): : This exact methodology operates primarily on arc-based formulations. Instead of pre-defining a high number of constraints, it begins with a relaxed version of the model and dynamically generates valid inequalities (cutting planes) at individual nodes of the search tree whenever fractional or invalid solutions occur. In rich VRP environments, these cuts typically include subtour elimination constraints (SECs), capacity cuts, and rounded capacity inequalities. This structural reinforcement tightens the dual bounds, significantly reducing the combinatorial search space before branching decisions are made (Pecin et al., 2017).
- Branch-and-Price (B&P): This framework implements the principle of Column Generation applied directly to route-based models, such as the Set Partitioning formulation (Feillet, 2010). Because the complete set of all mathematically feasible paths scales exponentially with the number of logistic vertices, standard solvers cannot load the entire matrix into memory. To circumvent this curse of dimensionality, B&P initializes a Restricted Master Problem (RMP) with a minimal subset of basic routes. It then iteratively evaluates the dual variables of the RMP and solves a pricing sub-problem (traditionally modeled as a Shortest Path Problem with Resource Constraints) to dynamically discover and inject profitable columns (routes with negative reduced costs) into the master formulation. Branching rules are subsequently enforced directly on the arcs when fractional path combinations arise.
- Branch-and-Cut-and-Price (BCP): Representing the current state-of-the-art computational boundary for exact solvability in vehicle routing, this hybridized framework unifies the strengths of both paradigms. BCP simultaneously manages the dynamic generation of columns (routes) via pricing sub-problems and the dynamic separation of rows (valid inequalities and cutting planes) to further tighten the convex hull of the master problem. By combining these advanced decomposition dual-bounds, BCP algorithms can establish mathematical optimality for highly constrained, heterogeneous routing instances up to roughly 100 customer nodes (Baldacci et al., 2008; Choi et al., 2020).

From an operational standpoint, understanding these exact frameworks highlights a critical threshold for decision support; while a standard multi-index MILP model can efficiently compute the global optimum for small-to-medium logistics networks, it faces a combinatorial wall when forced to resolve complex regional distributions, such as distributing multi-pallet demand from a central production hub to dozens of tightly constrained retail stores across the Ostwestfalen-Lippe (OWL) region. When commercial solvers encounter large-scale instances under strict time horizons, the computational burden of tracking individual asset indices causes the branch-and-bound tree to grow uncontrollably, underscoring the absolute necessity of metaheuristic alternatives to bridge the gap between theoretical optimality and operational feasibility. Consequently, this structural bottleneck provides the mathematical rationale for the empirical stress tests conducted in this study; as the graph size scales, the mathematical convergence of exact methods degrades, transitioning the tactical routing problem from an exhaustive combinatorial search to the specialized heuristic design detailed in the subsequent sections.

2.6 Heuristic and Approximate Approaches

Since the *HFVRPTW* belongs to the class of NP-hard formulations, exact solvers often fail to find solutions within reasonable operational timeframes when instances scale past 30–50 nodes (Cáceres-Cruz et al., 2014). To circumvent this, researchers utilize heuristic and metaheuristic frameworks designed to find high-quality, near-optimal solutions efficiently:

- Classical Heuristics (Constructive & Improvement) Modified Savings Frameworks: Modern adaptations of constructive algorithms alter the traditional "routing savings" calculations to dynamically incorporate heterogeneous capacity thresholds and fixed vehicle activation costs simultaneously (Coelho et al., 2016).
- Time-Oriented Insertion Heuristics: These algorithms sequentially insert unassigned customers into existing routes based on a weighted combination of spatial minimization and temporal delay minimization tailored for tight windows (Bräysy and Gendreau, 2005).
- Metaheuristics (Local Search Frameworks) Large Neighborhood Search (LNS) and Adaptive LNS (ALNS): Widely considered the gold standard for rich VRP variants (Pisinger and Ropke, 2007). ALNS works by iteratively destroying a portion of the current solution (removing customers using heuristics like random, worst, or radial removal) and rebuilding it using repair heuristics (Ropke and Pisinger, 2006). The algorithm adaptively learns which destroy/repair pairs perform best during the search.
- Variable Neighborhood Search (VNS) and Tabu Search: These methods navigate local search spaces by systematically changing neighborhood structures (e.g., swapping nodes between routes) while maintaining memory logs or structures to prevent the algorithm from getting trapped in local optima (Bräysy and Gendreau, 2005; Archetti et al., 2014).
- Hybrid Genetic Algorithms: Population-based metaheuristics that evolve a set of routing solutions over generations using advanced crossover and mutation operators, frequently hybridized with local search techniques to fix time window and capacity violations dynamically (Vidal et al., 2013).

The review of these heuristic and metaheuristic paradigms underscores a critical operational trade-off in mathematical programming: as regional distribution graphs expand, preserving computational tractability requires transitioning from exact convergence guarantees toward approximate solution methods. For large-scale logistics networks, where the exponential growth of the solution space renders exact commercial solvers inefficient within operational planning horizons, these metaheuristic frameworks provide the mathematical foundation necessary to bypass mathematical complexity. Consequently, the implementation of a custom-parameterized, approximate routing approach stands as the only viable mechanism to achieve high-quality, integer-feasible deployment schedules in negligible computational runtimes

2.7 Data Requirements for Model Implementation

To successfully operationalize the *Heterogeneous Fleet Vehicle Routing Problem with Time Windows (HFVRPTW)*, a comprehensive dataset capturing spatial, temporal, fleet, and demand characteristics must be systematically structured. Within the context of regional food distribution across the Ostwestfalen-Lippe (OWL) network (balancing multi-pallet delivery schedules from the central Dr. Oetker production facility (Node 0) to localized retail hubs (Lidl stores). The specific data parameters are formally categorized below:

Spatial and Network Infrastructure Data:

- Geographical Node Coordinates: Latitude and longitude coordinates for the central depot (i= 0) and all associated delivery vertices ($i \in \{1, \ldots, N\}$) to establish the spatial boundaries of the logistics network.
- Asymmetric Travel Distance Matrix: A definitive routing matrix mapping the shortest physical distance between each origin-destination vertex pair (i, j) traversing the regional road infrastructure (Cáceres-Cruz et al., 2014).
- Dynamic Travel Time Matrix: An asymmetric matrix tracking baseline transit times, augmented by time-dependent step-function congestion multiplier in order to capture commuter and freight delays (such as the A2 and A33 autobahns).

Customer Demand Characteristics:

- Deterministic Demand Volume: The discrete product volume requested by each retail location , standardized in pallet units to establish a direct volumetric coupling with vehicle load capacities (Koç et al., 2016).
- Spatial Accessibility Parameters: A binary infrastructure compatibility index denoting whether a specific vehicle is physically permitted to access and perform unloading operations at node j, accounting for urban bottlenecks, historical city centers, or loading bay dimensions.

Temporal Constraints and Service Parameters:

- Hard Customer Time Windows: Operational time interval bounding customer availability, defined by the earliest permissible arrival epoch and the latest allowable

service threshold. Any arrival prior induces a local waiting penalty, whereas any arrival post violates feasibility constraints (Bräysy & Gendreau, 2005).
- Composite Service Duration: Time dedicated to on-side operations at node j, mathematically structured as a function of a fixed administrative setup buffer and a variable discharge rate scaled by demand volume.

Heterogeneous Fleet Configurations

- Asset Fleet Allocation: The finite upper bound of active transport units available within each distinct vehicle category or size classification.
- Volumetric Load Capacity: The maximum payload capacity (quantified in standardized pallets) enforced as a hard upper bound to prevent asset over-allocation during route generation.
- Operational Cost Structure: A dual-component financial matrix consisting of a fixed asset activation cost triggered upon vehicle deployment, and a variable distance-based operational cost scaled per kilometer traversed to guide Gurobi's objective minimization function (Baldacci et al., 2008).
- Shift Horizon Thresholds: The maximum allowable duration for any individual route, serving as a legal and operational upper bound to enforce compliance with regional driving and rest-period labor frameworks.

2.8 Problem Simplifications and Operational Acceptability

To ensure the mathematical model developed in this project remains computationally feasible while still providing highly valuable decisions for the distribution networks of Dr. Oetker and Lidl, certain simplifications to the real-world problem were applied. These assumptions are explicitly stated and justified from both an operational supply chain perspective and a mathematical modeling standpoint:

Following established Operations Research literature, these assumptions are explicitly stated and justified from an operational and structural perspective:

- Data Availability & Approximation Note: Due to the highly sensitive and proprietary nature of corporate logistical data, commercial demand figures are not publicly available. To replicate a realistic environment, demand parameters were synthetically generated and calibrated based on node location (j), geographical density, and specific store sizes (footprint categorization). A comprehensive breakdown of this framework and its validation is further detailed in the *Data Collection section* of this report.
- Time-Dependent Parameterization of Travel Times: While travel times between logistics nodes are treated as deterministic during specific operational windows, the model avoids the oversimplification of static durations by embedding a time-dependent, piecewise step-function multiplier, calibrated for the Ostwestfalen-Lippe (OWL) road infrastructure. In real-world urban food distribution, assuming static travel times introduces severe time-window violation risks.

To avoid this, and given the lack of open-source, high-resolution historical traffic databases for the region, predictive AI modeling tools and advanced routing APIs were utilized to simulate and establish logical congestion ranges. These boundaries are heavily informed by documented bottleneck patterns on major regional arteries, specifically the A2 and A33 autobahn congestion patterns. To operationalize this mathematically, a dynamic **Traffic Congestion Factor** was developed to scale travel times across different operational windows. This factor provides the necessary adjustments to achieve highly accurate, time-sensitive durations, and its exact algorithmic structure and empirical calibration are further detailed in the *Data Collection* section of this report.

- Semi-Flexible Service and Unloading Times: Unloading and administrative service times at individual customer nodes are parameterized to establish a predictable, executable pattern within the mathematical algorithm. Large-scale retail actors operate under highly synchronized cross-docking environments where corporate logistics standards tightly govern incoming freight (Holzapfel et al., 2016). Dock scheduling slots are strictly regimented via time-slot management software.

  To balance algorithmic execution with real-world variance, the service time is mathematically disaggregated into two distinct components to maximize accuracy:

- Fixed Time Component: A static duration dedicated exclusively to inbound/outbound administrative driver paperwork and docking maneuvers.
- Variable Time Component: A linear function directly dependent on the specific volume (number of pallets) being unloaded from the designated vehicle size.

This hybrid structure ensures that while the solver benefits from structured deterministic patterns, the temporal constraints dynamically adjust to the delivery scale.

- Single-Period Planning Horizon and Driver Regulations

The vehicle routing optimization is executed for a single, discrete operational shift (a daily rolling horizon) without modeling multi-period vehicle re-allocation over an extended weekly horizon. High-frequency ambient and chilled food resupply networks operate on rigid daily turnaround cycles due to strict shelf-life requirements, cold-chain compliance, and retail inventory turnaround rates (Holzapfel et al., 2016). Optimizing on a single-period rolling horizon matches the true organizational decision-making framework of the dispatching teams without introducing the exponential variable growth and dimensionality curse typical of multi-period formulations.

- Operational Boundary Condition: In alignment with standard regional practice, the model assumes a single driver per vehicle policy per shift, omitting slip-seating or multi-driver slip operations. <u>However, to ensure legal and operational viability within the OWL region, strict European and German labor regulations regarding maximum allowable driving hours, mandatory break distributions, and total shift length</u>

constraints are explicitly embedded as upper-bound temporal limits within the route feasibility formulation.

- Fleet Loading Layout and Accessibility: Since the optimization model serves multiple customer nodes of varying sizes within a single route, it implicitly assumes that the spatial arrangement of pallets inside the vehicles allows for seamless unloading in the exact sequence of the routing schedule. To avoid the operational inefficiencies and delays associated with Last-In, First-Out (LIFO) obstructions (where cargo bound for subsequent stores blocks access to the current delivery) it is formally assumed that the loading dock layout and consolidation protocols at the central depot actively mitigate LIFO constraints. Freight is pre-sorted and reverse-loaded based on the optimized route sequence, ensuring immediate accessibility at each drop-off node.

- Depot Infrastructure, Emissions, and Speed Parameterization: To mathematically isolate the vehicle routing and capacity constraints, several boundary conditions regarding the physical fleet operations and environmental metrics are established:
    - Immediate Fleet Availability: In alignment with the cost parameters derived from official German logistics reports (BALM), the central depot in Bielefeld is assumed to possess a fully continuous, immediate availability of the required vehicle classes (7.5t, 18t, and 40t trucks) and the necessary dispatch infrastructure, eliminating exogenous queuing times or fuel-charging/refueling delays before departure.
    - Carbon Emissions Modeling: Environmental metrics, specifically the volume of CO_2 generated during distribution, are modeled strictly as a direct linear function of total mileage elapsed and vehicle-specific fuel consumption rates per truck category, omitting idle-engine emissions.
    - Velocity Mapping via Commercial Geographic Data: Vehicle travel speeds are constrained by the maximum legal speed limits authorized for each specific road category. Given the immense scale and density of the Ostwestfalen-Lippe (OWL) region road infrastructure, analyzing every individual street segment independently was computationally prohibitive. Consequently, standard velocity profiles and travel-time metrics were extracted directly from Google Maps API data engines to establish a robust, reliable, and realistic baseline for regional transit times.

- Single-Trip Vehicle Operations: The mathematical formulation operates under a strict single-trip constraint per vehicle within the active planning horizon. Each dispatched truck departs from the central depot in Bielefeld, executes its optimized sequence of customer deliveries, and returns directly to the depot to conclude its operational shift. The model explicitly omits multi-trip vehicle routing (MTVRP) mechanics; therefore, vehicles are not permitted to return to the central hub mid-day for replenishment, reloading, or subsequent re-dispatch to a secondary route.

- Forward Distribution (No Reverse Logistics)

The logistical network is modeled strictly as a forward distribution system where freight flows exclusively from the central depot to the retail customer nodes. To preserve model tractability and focus on outbound capacity utilization, reverse logistics operations are entirely omitted. The formulation assumes that no backhauls occur during route execution, meaning trucks do not collect empty pallets, damaged merchandise, or customer returns from the stores to transport them back to the Bielefeld depot.

- Fully Connected and Transitable Road Network

It is formally assumed that there are no unforeseen road closures, major traffic accidents, construction detours, or extreme weather events that would completely compromise network connectivity. Furthermore, the model assumes that standard designated freight corridors are fully accessible, omitting micro-level urban transit bans.

- Deterministic and Uniform Fuel Cost Parameters

Operating cost metrics across the fleet are assumed to be deterministic and directly proportional to the vehicle category (7.5t, 18t, and 40t trucks), applying fixed rates of €0.25/km, €0.38/km, and €0.56/km respectively. The model assumes a uniform fuel price environment throughout the operational shift, omitting real-time intra-day fluctuations at retail refueling stations or regional variance in energy pricing across the OWL territory.

III. Data Collection

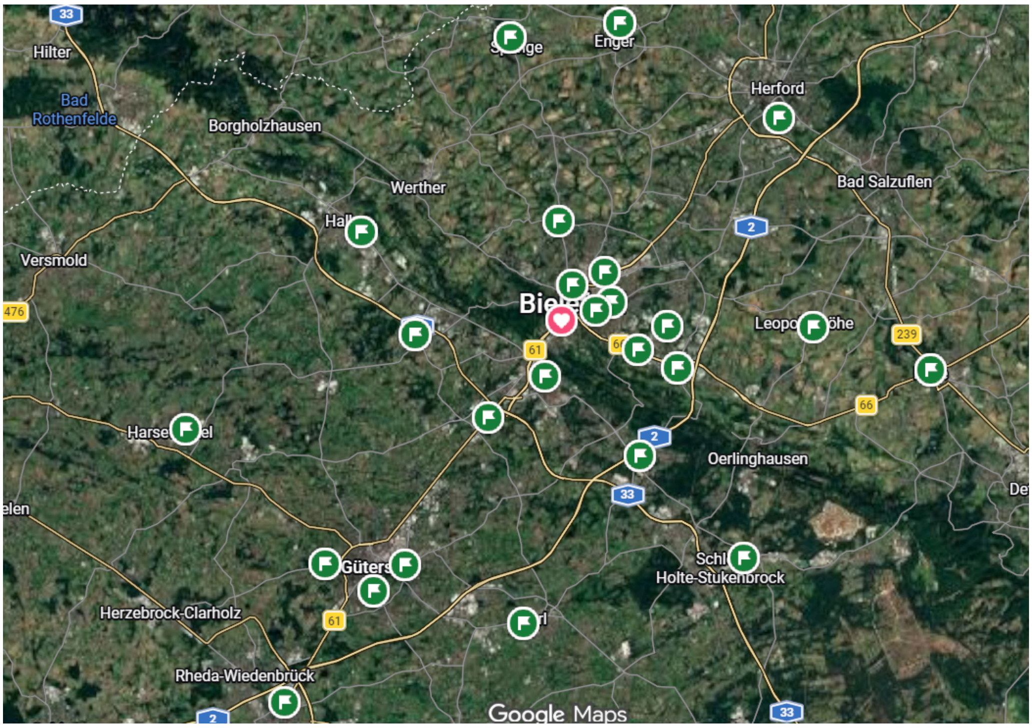

Figure 1: Google Maps visualization of the entire spatial distribution of the logistics network across the OWL region.

To understand the spatial constraints governing the routing optimization, the logistics network was mapped using Google Maps. The layout illustrates the exact positioning of the central Dr. Oetker supply hub (depot) and the 25 decentralized Lidl delivery points.

This visualization is highly valuable for decision-support analysis, particularly when determining fleet allocation policies: it highlights spatial sub-regions where multiple retail nodes are clustered close together, providing a clear justification for why high-tonnage multi-axle trucks are appropriate for specific industrial or open road corridors. Conversely, it shows isolated or dense central nodes where strict structural and environmental bottlenecks physically limit operations to low-capacity vehicles.

1. Distance and Time matrixes (table 1 and table 2)

To mathematically operationalize the routing network, the spatial and temporal friction between nodes is captured through two fundamental asymmetric matrices: the Distance Matrix and the Travel Time Matrix.

The Distance Matrix, measured strictly in kilometers, maps the physical separation from any origin node i to any destination node j across the 25 operational nodes (comprising the Dr. Oetker central depot and the 24 retail Lidl stores). Complementing this, the Travel Time Matrix translates these spatial intervals into temporal dimensions measured in minutes. Rather than assuming static, idealized velocities, this temporal parameter accounts for regional speed limits and baseline urban friction inherent to the road topology. Both datasets were empirically retrieved utilizing Google Maps API services, providing a highly reliable and realistic deterministic framework that allows the optimization solver to identify mathematically sound sequences and establish the optimal dispatch scheduling."

i/j	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
0	0	13	14	9	16	16	15	16	13	9	21	9	28	26	30	25	27	32	25	20	17	29	31	33	30	32
1	13	0	16	7	12	20	19	15	8	18	21	10	31	29	34	29	31	34	28	23	20	24	32	31	28	30
2	14	16	0	13	20	20	22	23	17	11	18	17	26	24	28	23	25	28	23	18	14	25	23	31	27	27
3	9	7	13	0	9	18	17	17	10	13	16	11	31	29	33	28	30	33	28	23	20	21	29	29	24	27
4	16	12	20	9	0	27	26	22	18	22	26	19	31	28	32	28	30	33	27	23	19	21	25	29	22	22
5	16	20	20	18	27	0	6	7	15	16	10	15	27	25	33	26	26	37	22	27	24	19	38	22	18	32
6	15	19	22	17	26	6	0	6	10	13	15	10	35	33	37	33	35	35	32	21	21	29	17	37	24	22
7	16	15	23	17	22	7	6	0	8	16	14	10	30	24	28	24	27	29	24	21	18	29	35	33	28	36
8	13	8	17	10	18	15	10	8	0	18	21	5	25	24	31	25	24	36	20	26	22	13	35	20	17	31
9	9	18	11	13	22	16	13	16	18	0	14	15	23	22	29	23	22	33	17	23	20	15	38	18	15	29
10	21	21	18	16	26	10	15	14	21	14	0	19	20	15	19	20	24	23	22	17	13	28	35	31	27	37
11	9	10	17	11	19	15	10	10	5	15	19	0	20	18	26	18	19	30	13	20	17	19	39	22	18	32
12	28	31	26	31	31	27	35	30	25	23	20	20	0	9	10	18	14	24	29	32	25	34	48	37	33	47
13	26	29	24	29	28	25	33	24	24	22	15	18	9	0	12	16	16	22	27	29	20	31	47	35	31	45
14	30	34	28	33	32	33	37	28	31	29	19	26	10	12	0	25	14	18	35	28	23	40	45	45	40	51
15	25	29	23	28	28	26	33	24	25	23	20	18	18	16	25	0	19	34	22	28	25	33	47	37	32	46
16	27	31	25	30	30	26	35	27	24	22	24	19	14	16	14	19	0	25	27	30	27	32	50	36	31	45
17	32	34	28	33	33	37	35	29	36	33	23	30	24	22	18	34	25	0	37	20	17	43	35	49	42	43
18	25	28	23	28	27	22	32	24	20	17	22	13	29	27	35	22	27	37	0	26	23	20	43	25	26	41
19	20	23	18	23	23	27	21	21	26	23	17	20	32	29	28	28	30	20	26	0	10	34	24	40	33	29
20	17	20	14	20	19	24	21	18	22	20	13	17	25	20	23	25	27	17	23	10	0	28	28	30	27	33
21	29	24	25	21	21	19	29	29	13	15	28	19	34	31	40	33	32	43	20	34	28	0	38	12	19	31
22	31	32	23	29	25	38	17	35	35	38	35	39	48	47	45	47	50	35	43	24	28	38	0	39	24	12
23	33	31	31	29	29	22	37	33	20	18	31	22	37	35	45	37	36	49	25	40	30	12	39	0	20	33
24	30	28	27	24	22	18	24	28	17	15	27	18	33	31	40	32	31	42	26	33	27	19	24	20	0	17
25	32	30	27	27	22	32	22	36	31	29	37	32	47	45	51	46	45	43	41	29	33	31	12	33	17	0

Figure 2. Distance matrix (km) for the 25 nodes within the scope.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA
i/j	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
0	0	5.1	6.5	2.9	6.3	7.1	7.1	7	3.7	3.7	10.7	2.5	28.5	17.1	20.2	21.1	34	23	23.7	14.5	11.2	16.5	16.5	21.6	23.4	20.8
1	5.1	0	11.3	2.9	4.9	7.9	7.8	6	2.1	8.9	19.9	3.1	29.3	18.3	21.3	21.9	35.2	34.2	24.4	15.3	12.4	13	15.9	20.7	17.2	19.4
2	6.5	11.3	0	9.2	12.6	20.8	11.8	13.5	12.2	4.5	10.9	8.9	29.1	17.8	20.8	21.7	34.6	33.6	24.3	15.1	11.8	14.7	13.7	22.7	17.4	18.8
3	2.9	2.9	9.2	0	3.5	8.9	8.9	8.9	3.5	6.4	17.4	2.3	31.7	20.3	23.3	24.3	37.2	36.2	26.9	17.7	14.4	12.1	16.6	20.4	16	19.2
4	6.3	4.9	12.6	3.5	0	12.3	12.3	11	6.8	9.8	20.8	6.4	31.6	20.2	23.2	24.2	37.1	36.1	26.8	17.6	14.3	12.7	14.8	20.6	14.8	17.3
5	7.1	7.9	20.8	8.9	12.3	0	2.4	3.7	6.6	9.8	7.2	6.1	27.9	26.7	31.5	27.6	33.4	36.2	16.5	17.8	14.5	13.3	19.7	17.4	19.2	32.8
6	7.1	7.8	11.8	8.9	12.3	2.4	0	2.6	4.2	6.7	10.2	3.7	32.5	21.1	24.2	25.1	38	30	27.7	15.9	13.7	15.6	10.7	25.8	16.9	13.7
7	7	6	13.5	8.9	11	3.7	2.6	0	3.8	8.8	11.2	4.2	16.2	13.8	16.9	15.1	31.2	21.4	15	13.6	10.3	19	21.5	23.3	28.7	25.7
8	3.7	2.1	12.2	3.5	6.8	6.6	4.2	3.8	0	8.5	11.7	1.4	27.2	25.9	30.7	26.9	32.7	37.6	15.8	31.2	24.6	8.8	20.6	16.7	18.5	32
9	3.7	8.9	4.5	6.4	9.8	9.8	6.7	8.8	8.5	0	7.5	5.3	26.1	24.9	29.7	25.8	31.7	36.5	12.7	30.2	23.6	9.6	23.7	15.7	17.5	31
10	10.7	19.9	10.9	17.4	20.8	7.2	10.2	11.2	11.7	7.5	0	11.1	12.6	9.2	12.2	13.1	30.1	17.4	19.8	16.5	7.7	27.6	22.8	31.7	33.3	27.1
11	2.5	3.1	8.9	2.3	6.4	6.1	3.7	4.2	1.4	5.3	11.1	0	19	17.8	22.6	12.6	24.6	35.6	8.5	23.1	16.5	17.1	30.8	21.2	22.8	36.3
12	28.5	29.3	29.1	31.7	31.6	27.9	32.5	16.2	27.2	26.1	12.6	19	0	3.1	4.6	9.5	8.7	17.1	26.2	22	15.6	33.6	37.9	37.7	39.3	51
13	17.1	18.3	17.8	20.3	20.2	26.7	21.1	13.8	25.9	24.9	9.2	17.8	3.1	0	5.7	8.2	10.4	15.6	24.9	20.2	13.4	32.3	31.7	36.5	38	49.7
14	20.2	21.3	20.8	23.3	23.2	31.5	24.2	16.9	30.7	29.7	12.2	22.6	4.6	5.7	0	13	8.7	12.8	29.8	21.4	16.2	37.2	37.1	41.3	42.9	41.2
15	21.1	21.9	21.7	24.3	24.2	27.6	25.1	15.1	26.9	25.8	13.1	12.6	9.5	8.2	13	0	14.3	24.2	13.2	27.7	21.1	29.1	35.5	33.3	34.9	46.5
16	34	35.2	34.6	37.2	37.1	33.4	38	31.2	32.7	31.7	30.1	24.6	8.7	10.4	8.7	14.3	0	20.5	31.1	41.1	34.5	38.5	48.9	42.6	44.2	55.9
17	23	34.2	33.6	36.2	36.1	36.2	30	21.4	37.6	36.5	17.4	35.6	17.1	15.6	12.8	24.2	20.5	0	41.9	17.1	13.8	49.7	31.7	53.8	55.4	35.6
18	23.7	24.4	24.3	26.9	26.8	16.5	27.7	15	15.8	12.7	19.8	8.5	26.2	24.9	29.8	13.2	31.1	41.9	0	29.3	22.7	13.9	47.7	17.9	27.6	39.2
19	14.5	15.3	15.1	17.7	17.6	17.8	15.9	13.6	31.2	30.2	16.5	23.1	22	20.2	21.4	27.7	41.1	17.1	29.3	0	6.9	38.1	17.3	42.2	43.8	21.4
20	11.2	12.4	11.8	14.4	14.3	14.5	13.7	10.3	24.6	23.6	7.7	16.5	15.6	13.4	16.2	21.1	34.5	13.8	22.7	6.9	0	31.3	21	35.5	37	25.4
21	16.5	13	14.7	12.1	12.7	13.3	15.6	19	8.8	9.6	27.6	17.1	33.6	32.3	37.2	29.1	38.5	49.7	13.9	38.1	31.3	0	25.1	8.2	14.2	27.8
22	16.5	15.9	13.7	16.6	14.8	19.7	10.7	21.5	20.6	23.7	22.8	30.8	37.9	31.7	37.1	35.5	48.9	31.7	47.7	17.3	21	25.1	0	32.5	18.7	7.5
23	21.6	20.7	22.7	20.4	20.6	17.4	25.8	23.3	16.7	15.7	31.7	21.2	37.7	36.5	41.3	33.3	42.6	53.8	17.9	42.2	35.5	8.2	32.5	0	16	29.6
24	23.4	17.2	17.4	16	14.8	19.2	16.9	28.7	18.5	17.5	33.3	22.8	39.3	38	42.9	34.9	44.2	55.4	27.6	43.8	37	14.2	18.7	16	0	15.9
25	20.8	19.4	18.8	19.2	17.3	32.8	13.7	25.7	32	31	27.1	36.3	51	49.7	41.2	46.5	55.9	35.6	39.2	21.4	25.4	27.8	7.5	29.6	15.9	0

Figure 3. Time matrix (minutes) for the 25 nodes within the scope.

2. Store Classification and Parameterization Logic

The physical category size of a store (determined by its geographic location and urban zoning) directly impacts its backroom storage capacity, the turnover of ambient and frozen products, and (crucially) the type of truck that can access the facility. If a store is located in a narrow historical district where only a small truck (Low) can fit, its physical demand cannot exceed that truck's maximum capacity. Otherwise, the MIP model would become infeasible unless multiple delivery visits were permitted.

Store Categories

- Group A: Small / Express Stores (Historical Districts / Dense Urban Centers): Limited space, high turnover, but replenished in small batches.
  - *Estimated Demand:* 2 - 5
- Group B: Medium / Standard Stores (Residential Areas / Connector Streets): Average-sized neighborhood Lidl supermarkets.
  - *Estimated Demand:* 6 - 12

- Group C: Large / Hyper-Lidl or Hub Stores (Industrial Zones / Open Highways / Arterial Roads): Large suburban commercial complexes equipped with full loading docks and high storage capacity.
  - *Estimated Demand:* 14 - 25

3. **Demand Parameter Profiles & Data Distribution Model (d_j)**

The baseline demand is modeled using the following function based on the store's microlocation characteristics:

$$d_j = \textbf{Footprint Base} + \textbf{Transit Density Bonus} - \textbf{Spatial Buffer}$$

To make the formula work, we establish fixed numerical values for the qualitative features of each location category:

1. *Footprint Base:* Tied directly to the store category size.
   - Small = 3 pallets (Compact retail square footage, limited backroom cold-storage footprint)
   - Medium = 8 pallets (Standard standalone suburban supermarket layout)
   - Large = 18 pallets (Regional flagship hub with extensive warehouse facilities).
2. *Transit Density Bonus*: Added if the store sits on a high-traffic highway, main commuter street, or commercial corridor (+1 to 6 pallets depending on regional prominence).

   +0: Minor neighborhood road or restricted town pedestrian zone.

   +1: Standard multi-lane city commuter street.

   +2/+3: Major arterial cross-town ring road or federal highway ($B\text{-Road}$).

   +4/+6: Direct accessibility to a major federal Autobahn highway interchange ($A2 / A33$).

3. *Spatial Buffer:* Subtracted if the store has severe physical restrictions, localized competition, or if the demand is splits in a lower-density municipal zone (-1 to -3 pallets).

   0: No limiting spatial factors; captured demand is fully sustained.

   -1: Minor spatial restrictions (example: historical core layout slows turnaround)

   -2/-3: Shared municipal market or adjacent direct regional grocery competitors splitting the shopping catchment basin.

This rigorous, systematic translation converts qualitative spatial characteristics of the Ostwestfalen-Lippe retail landscape directly into fixed, reproducible numerical constraints. This enables the model to bypass corporate confidentiality restrictions while providing an auditable data model for the optimization solver.

4. **Lidl Store Size Classification Framework**

- Small=S
    - Average Sales Area: 600 m^2 - 900 m^2
    - Backroom Warehouse Size (Approximated) = 100m^2 - 150 m^2
    - Rack Capacity: 2–3 standard industrial rack levels high, strictly limited to a total of 15–25 pallet slots in backroom storage (due to architectural or historical constraints, narrow layouts, or structural building columns).
    - Operational Demand Range: 2 to 5 pallets per delivery
    - Logistical Rationale: These stores rely on a *Just-In-Time* continuous replenishment loop. Their compact backroom freezers cannot accept bulk freight without causing floor congestion. Physically, they are restricted to low-capacity delivery vehicles (Low fleet only) due to tight inner-city geometric constraints.
- Medium = M
    - Average Sales Area: 1000 m^2 - 1400 m^2
    - Backroom Warehouse Size (Approximated) = 200m^2 - 300 m^2
    - Rack Capacity: 3–4 heavy-duty industrial pallet rack levels high, storing 60–100 pallet slots overall.
    - Operational Deamdn Range: 2 to 5 pallets per delivery

- Logistical Rationale: This represents the classic suburban standalone supermarket with its own dedicated surface parking lot and loading area. The warehouse size is optimized to cleanly absorb a standard mixed ambient/cold-chain delivery drop from a medium-sized vehicle without disturbing ongoing stock rotation.

- Large= L
  - Average Sales Area: 1500 m^2 - 2,200 m^2
  - Backroom Warehouse Size (Approximated) = 400m^2 - 600 m^2
  - Rack Capacity: 4–5 multi-tier industrial storage racks with dedicated drive-in bays, allowing **150–250+ pallet slots** in backroom inventory.
  - Operational Demand Range: 14 to 25 pallets per delivery.
  - Logistical Rationale: High-volume peripheral branches strategically located near highways, arterial loops, or light industrial commercial clusters. They are built with fully automated docking ramps and heavy freight loops designed to handle massive, multi-axle heavy-duty trucks loaded to maximum capacity (High fleet permitted).

This physical classification ensures that generated data parameters correspond directly with fleet routing capacities, preventing the mathematical solver from assigning unfeasible bulk shipments to structurally restricted nodes.

5. Pallet demand for each store

Node	Location	Road Zone Type	Size Classification	Generated demand (pallets) (di)	Technical Justification
N_1	Teutoburger Straße 98, 33607 Bielefeld-Mitte	Historic / Central urban	S	FootPrint: 3 Transit Density Bonus: 0 (Pedestrian Zone) Spatial Buffer: 0 d1= 3 pallets	Positioned deep within an urban neighborhood center. The compact city square footage prevents high pallet stock intake, and because deliveries must happen through a restricted courtyard, the transit bonus is 0.
N_2	Jöllenbecker Straße 41, 33613 Bielefeld-Mitte	Historic / Central urban	S	FootPrint: 3 Transit Density Bonus: 1 (Commuter Street Bonus) Spatial Buffer: 0 d2 = 4 pallets	Although classified as a small store footprint, it sits on a primary high-density city commuter street out of the central ring, which awards a +1 transit bonus. Rapid shelf-depletion rates drive up the demand by 1 extra pallet compared to N1.

N_3	Heeper Straße 113, 33607 Bielefeld-Mitte	Residential zone	M	FootPrint: 8 Transit Density Bonus: 0 (Standard Access) Spatial Buffer:0 d3 = 8 pallets	A classic suburban standalone Lidl supermarket, capturing a standard middle-class housing development demographic. Perfect operational baseline without external density shocks.
N_4	Stadtheider Straße 11, 33609 Bielefeld-Mitte	Industrial zone	L	FootPrint: 18 Transit Density Bonus: 0 (Industrial Baseline) Spatial Buffer:0 d4= 18 pallets	A large-scale branch built inside a commercial/light industrial inner-city pocket.
N_5	Detmolder Straße 345, 33605 Bielefeld-Stieghorst	Open / Arterial road	L	FootPrint: 18 Transit Density Bonus: 4 (Autobahn Entrance Corridor Bonus) Spatial Buffer: 0 d5 = 22 pallets	Sits squarely on the main arterial route leading out toward the B66 and A2 Autobahn entry points. Exceptional transit flow generates massive multi-pallet replenishment requirements (+4 bonus).
N_6	Babenhauser Straße 18, 33613 Bielefeld-Schildesche	Residential zone	M	FootPrint: 8 Transit Density Bonus: 0 (Student demographics buffer) Spatial Buffer: -1 7 pallets	Close to university residential sectors. While customer volume is high, the purchase profile leans toward single-person baskets and smaller packaging sizes rather than massive multi-pallet family stocks, triggering a small operational buffer reduction of -1
N_7	Kimbernstraße 1, 33647 Bielefeld-Brackwede	Open road	L	FootPrint: 18 Transit Density Bonus: 0 Spatial Buffer: -2 d7 = 16 pallets	An expansive retail park location. However, severe morning rush-hour congestion along the industrial access corridors slightly chokes rapid early-day truck turnarounds, applying a -2 spatial buffer adjustment.
N_8	Oldentruper Straße 255, 33719 Bielefeld-Heepen	Industrial zone	L	FootPrint: 18 Transit Density Bonus: 2 (Commercial Hub Axis) Spatial Buffer:0 d8 = 20 pallets	Embedded into a heavy industrial-commercial axis with flawless multi-lane access roads that support non-stop high-capacity freight distribution (+2 bonus).
N_9	Detmolder Straße 550, 33699 Bielefeld-Stieghorst	Open / Arterial road	L	FootPrint: 18 Transit Density Bonus: 6 Spatial Buffer:0 d9 = 24 pallets	The absolute flagship store of the network, resting directly at the highway interchange border. It operates as a regional cross-dock shopper hub, earning the maximum +6 transit infrastructure bonus.

N_{10}	Kasseler Straße 1, 33649 Bielefeld-Brackwede	Open road	M	FootPrint: 18 Transit Density Bonus: 0 Spatial Buffer: 3 (Heavy Local congestion buffer) d10 = 15 pallets	While it is a huge retail facility, it sits on a major southern exit bottleneck that suffers from extreme rush-hour local gridlock, forcing a -3 operational spatial buffer to account for delayed delivery processing.
N_{11}	Henleinstraße 1, 33689 Bielefeld	Industrial zone	L	FootPrint: 18 Transit Density Bonus: 1 (Industrial Park Collector) Spatial Buffer: 0 d11 = 19 pallets	Positioned inside a spacious modern industrial park. Built explicitly for heavy freight turnover with clean geometric docking bays (+1 bonus).
N_{12}	Neuenkirchener Str. 100, 33332 Gütersloh	Open road	M	FootPrint: 8 Transit Density Bonus: 3 (Cross-Town Highway Bonus) Spatial Buffer: 0 d12 = 11 pallets	Situated on a high-speed arterial cross-town highway in Gütersloh. It captures high-volume suburban car traffic and commuter flows heading back toward outer residential sectors (+3 bonus).
N_{13}	Carl-Bertelsmann-Straße 152, 33332 Gütersloh	Industrial zone	L	FootPrint: 18 Transit Density Bonus: 3 Corporate Headquarters Corridor) Spatial Buffer: 0 d13 = 21 pallets	Directly adjacent to massive multi-national corporate headquarters and manufacturing campuses in Gütersloh. High worker density creates a huge local consumer pool (+3 bonus).
N_{14}	Auf dem Stempel 10, 33334 Gütersloh	Residential zone	M	FootPrint: 8 Transit Density Bonus: +1 (Residential Collector Road) Spatial Buffer: 0 d14 = 9 pallets	Embedded directly into a massive family housing development. High average cart volumes for bulk-packed household goods justify a minor +1 transit density flow increase.
N_{15}	Westring 11, 33415 Verl	Open road	L	FootPrint: 18 Transit Density Bonus: 0 Spatial Buffer: -1 (Peripheral Delay Buffer) d15= 17 pallets	Located on Verl's outer high-speed transit ring. It has a spacious footprint, but its remote peripheral distance adds slight transit variances, represented by a small -1 buffer constraint.
N_{16}	Bielefelder Str. 55, 33378 Rheda-Wiedenbrück	Historic/ Central urban	S	FootPrint: 3 Transit Density Bonus: 0 (Standard Core) Spatial Buffer:0 d16 = 3 pallets	Serves a historical smaller city core with lower population density than Bielefeld-Mitte, yielding a steady, basic baseline without local traffic or competition distortions.

N_{17}	Berliner Ring 40, 33428 Harsewinkel	Open road	M	FootPrint: 8 Transit Density Bonus: 4 (Major Regional Byoass Bonus) Spatial Buffer: 0 d17 = 12 pallets	Located on the primary regional bypass ring road. This location captures heavy rural automobile traffic entering and exiting Harsewinkel, justifying a significant +4 transit density increase.
N_{18}	Lüchtenstraße 61, 33758 Schloß Holte-Stukenb rock	Residenti al zone	M	FootPrint: 8 Transit Density Bonus: 0 Spatial Buffer: -2 (Regional Competitor Split Buffer) d18 = 6 pallets	This node operates in a lower-density municipal zone with close, direct competition from overlapping regional supermarkets on the same street, resulting in a -2 demand split buffer.
N_{19}	Elsa-Brändströ m-Straße 1, 33790 Halle (Westfalen)	Residenti al zone	M	FootPrint: 8 Transit Density Bonus: 0 Spatial Buffer:0 d19 = 8 pallets	Positioned optimally next to public sports facilities and residential blocks, preserving a perfectly clean, uninterrupted medium-size store baseline demand.
N_{20}	Bahnhofstraße 23, 33803 Steinhagen	Historic / Central urban	S	FootPrint: 3 Transit Density Bonus: 1 (High Street Flow) Spatial Buffer: 0 d20 = 4 pallets	Located on Steinhagen's main retail high street. Heavy transit passenger flow from neighboring transit lines guarantees a steady inventory drawdown, prompting a +1 transit adjustment.
N_{21}	Krentruper Str. 34, 33818 Leopoldshöhe	Open road	M	FootPrint: 8 Transit Density Bonus: 2 (Commuter Town Feeder) Spatial Buffer: 0 d21 = 10 pallets	Serves an expanding commuter town between Bielefeld and Lemgo. High home-ownership rates mean larger family grocery baskets, warranting a +2 logistical flow bonus.
N_{22}	Lange Str. 35-37, 32139 Spenge	Historic / Central urban	S	FootPrint: 3 Transit Density Bonus: 2 (Regional Feeder Bonus) Spatial Buffer:0 d22= 5 pallets	Spenge is a peripheral town core. This store acts as a primary grocery point for multiple surrounding rural settlements, capturing out-of-town car commuters on the main municipal feeder road (+2 bonus).
N_{23}	Friedrich-Petri -Straße 18-26, 32791 Lage	Residenti al zone	M	FootPrint: 8 Transit Density Bonus: 1 (Municipal Thoroughfare) Spatial Buffer:0 d23 = 9 pallets	A standard medium municipal branch in the Lippe district capturing regular daily vehicle movements along the city's connecting thoroughfare (+1 bonus).
N_{24}	Ahmser Str. 50-54, 32052 Herford	Industrial zone	L	FootPrint: 18 Transit Density Bonus: 5 (Industrial Corridor Master Axis) Spatial Buffer: 0	A massive flagship location anchoring Herford's main industrial logistics corridor. Its extensive cold-chain bays allow it to handle heavy bulk loads seamlessly, yielding a +5 infrastructure bonus.

				d24 = 23 pallets	
N_{25}	Lambernweg 46, 32130 Enger	Open Road	M	FootPrint: 8 Transit Density Bonus: 2 (Outer Ring Capture) Spatial Buffer:0 d25 = 10 pallets	Positioned directly on the outer connecting ring road, intercepting peripheral rural traffic prior to entry into Enger's retail center (+2 bonus).

Table 1. Demand Generation and Node Profiles for the 25 Network Locations

6. Fleet Characterization: Fuel Profiles, Market Pricing

To operationalize the objective function (specifically balancing total operating costs against environmental impacts) the heterogeneous fleet must be baseline-parameterized regarding its energy source, current fuel expenditures, and carbon footprints.

- Fuel Usage and Local Pricing Matrix

In commercial German freight transport, standard internal combustion engine (ICE) assets rely almost exclusively on automotive diesel fuel (DIN EN 590). According to data from the German Federal Statistical Office (Statistisches Bundesamt - Destatis, 2026) and the European Commission's Weekly Oil Bulletin, retail diesel pricing in Germany sits at approximately €1.76 to €1.84 per liter (Destatis, 2026). For corporate logistics networks executing private-fleet distribution from a central point like the Dr. Oetker plant in Bielefeld, bulk commercial procurement contracts traditionally discount this standard pump price by roughly 10–15%, yielding a baseline operational fuel cost of €1.55 per liter.

- Traffic congestion security factor

The Ostwestfalen-Lippe (OWL) region, centered around Bielefeld, Gütersloh, and Herford, represents one of the most dense economic and industrial clusters in Germany. It serves as a primary logistical corridor connecting western German manufacturing hubs with trans-European freight routes via the A2 and A33 autobahns. Therefore, this network suffers from severe asymmetry: heavy freight transport coexists with intense traffic along critical bottlenecks (such as Ostwestfalenstraße, Detmolder Straße, Herford Straße) and key autobahn interchanges like the Kreuz Bielefeld (A2/A33).

In traditional OR, travel times are often assumed to be fixed, although in a real-world scenario like food distribution from the central Dr. Oetker plant, assuming static travel times introduces major scheduling risks. If a vehicle departs during peak commuter hours, ignoring traffic delays leads to time-window violations, resulting in failed deliveries, administrative penalties, and spoiled cold-chain products. Consequently, to better represent commuter and freight congestion patterns on the Bielefeld and wider OWL road network, it was established a “buffer constraint” penalizing or favoring travel times based on the exact departure epoch from the origin vertex:

7. Key Empirical and Operational Rationales for congestion multiplier

Congestion Multiplier μ (t) =	*Temporal Framework*	*Regional Infrastructure Stress*	*Urban Network Validation*
1.25 ~1.35 Peak hours	Morning Rush: 07:30 ≤ t < 09:00 Late afternoon, industrial shift-changes: 16:00 ≤ t < 18:00	According to the German *Federal Highway Research Institute*, the A2 autobahn and B61 corridor passing through the Bielefeld-Gütersloh-Herford economic hub display severe micro-capacity constraints during these intervals due to overlapping freight logistics and private transit.	Data extracted from the TomTom Traffic Index (Bielefeld Cluster) indicates that travel-time delays during these peak windows surge by 30% to 40% compared to free-flow baselines. According to the German Federal Highway Research Institute (BASt), these corridors operate at near-saturation levels during industrial shift changes.
1~ 1.15 (Mid-Day Window)	Standardized operational mid-day window 09:00 ≤ t < 16:00	During these hours, regional traffic flow stabilizes back to expected fluid conditions. Travel time calculation is determined strictly by the geographic distance with the static admin. safety buffer.	Representative of steady-state network conditions. Base travel times are calculated utilizing the actual underlying road network topology (real street-routing distances). This congestion multiplier serves as an operational buffer to incorporate delays to physical infrastructure (for example: traffic light cycles, or municipal speed limits).
≤1 Free-Flow	Night and Early Morning: t ≤ 7:30 t ≤ 18:00	In food-supply distribution networks (e.g., pallet delivery to regional Lidl outlets), heavy-duty fleet operations rely heavily on early hours to exploit clear road infrastructure and bypass urban transit bans.	Food-supply distribution networks heavily exploit this early-morning window to optimize heavy-duty fleet deployment, ensuring that high-capacity trucks clear central urban links before municipal delivery bans activate.

Table 2. Traffic congestion factor μ (t) depending on the temporal framework, regional infrastructure stress and network justification.

8. Vehicle capacity limitations

Category	Features	Vehicle type	Dimensions (L x W x H)
Low	• Small vehicle. • Low fixed cost. • High maneuverability. • Suitable for stores with low demand.	Motor Vehicle (7.5 t)	6.00 x 2.45 x 2.40 m

Medium	• Typical regional distribution vehicle. • Good balance between capacity and accessibility.	Trailer (18 t)	8.10 x 2.45 x 2.5 m
High	• European standard vehicle with maximum capacity. • Maximum load consolidation. • Lower cost per pallet transported. • Low Maneuverability: This type cannot access some streets due to its size.	Articulated truck (40 t)	13.6 x 2.45 x 2.70 m

Table 3. Vehicle characteristics for each type of category.

Standard Euro Pallet size considered: $80\ cm \times 120\ cm$ (EN 13698 - 1:2003).
Any solution violating these conditions is considered infeasible.

Capacity measure	Low	Medium	High
Euro pallets (Q_k)	15	20	34
Payload (t)	2.8	13.5	25

Table 4. Vehicle capacity in terms of Europallets and payload for each category type.

Dimensions, payloads, and Europallets retrieved from: (Orbiet Logistik, 2021), considering the Council Directive 96/53/EC.

These specific vehicles were chosen because they exhibit significant fluctuations in an optimization study, allowing the algorithm to select the ideal option across different vehicle sizes. This selection represents a clear leap in capability and reflects distinctly differentiated operational levels.

Example (in terms of euro pallets capacity):

$$Medium - Low: 20 - 15 = 5\,pallets: 5/15: \ + 33.333\%$$
$$High - Medium: 34 - 20 = 14\,pallets: 14/20: \ + 70\%$$
$$High - Low: 34 - 15 = 19\,pallets: 19/15: \ + 126.667\%$$

Far enough apart for the model to reveal differences in:

- Number of routes
- Vehicle utilization
- Adherence to time windows
- Total cost
- Total operating time

The size truck- capacity (Q) is different, $Q_{low} \neq Q_{Medium} \neq Q_{High}$ and affects the next factors directly:

- Gas cost consumption in euros (C): $C_{low} \neq C_{Medium} \neq C_{High}$
- Service time restrictions (D): $D_{low} \neq D_{Medium} \neq D_{High}$

Factors	Low	Medium	High
C (euros)	~ 0.25 € / km	~ 0.38 € / km	~ 0.56 € / km
D (hours)	6	8	9

Table 5. Classification of Cost (C) and Service Time (D Restrictions by Capacity

Factor C (retrieved from *Verbraucherpreisindex für Deutschland*):

The baseline fuel consumption costs for the vehicle categories (€0.25/km for the Low category, €0.38/km for the Medium category, and €0.56/km for the High category) are derived from operational data published by the German Federal Office for Logistics and Mobility in its periodic Verbraucherpreisindex für Deutschland reports.

These values are calculated by crossing the industry-standard fuel consumption rates for European fleets (averaging 14L, 22L, and 33L per 100 km, respectively) with the net commercial diesel prices in Germany, as reported by the Federal Statistical Office (Destatis).

These parameters provide a realistic baseline for the Rich Vehicle Routing Problem (RVRP) objective function, as they reflect actual operating conditions and fuel price standards within the German freight transport sector.

Factor D and V:

Service time restrictions (6, 8, and 9 hours) are aligned with the daily driving limits established under Regulation (EC) No. 561/2006. Likewise, the maximum number of stores per vehicle type (6, 10, and 12, respectively) is not an arbitrary parameter, but an operational constraint derived from time capacity:

Standard Activity Time (T_{total})

For this optimization model, the total time for each store (T_{total}) $in\ minutes$ is composed of a fixed time, independent of the vehicle size, and a variable time depending on the pallet volume that it is carrying.

Sequence	Activity	Estimated Net time (minutes)	Margin of error (minutes)	(T_{total})	Operative justification
1	Parking and Maneuvering	10	+5 (yard traffic/waiting)	15	Time from when the truck crosses the gate until it is perfectly docked at the loading bay.
2	Paperwork and Opening	5	+5 (Searching for the supervisor)	10	Breaking security seals, the supervisor's signature is on the delivery note.
3	Physical Unloading	1 p/pallet	+0.5 (stuck/misaligned pallet)	1.5	Use of an electric pallet jack.
4	Empty Pallet Collection	0.5 p/pallet	+0.5 (Pallet inspection)	1	Loading empty pallets.

Table 6. Breakdown of standard activity times, including margins of error in minutes.

Service time (T_{si})	Activity description	Example	Time (minutes)
Variable time $(t_{Descharge} + t_{inspection})$: (1.5 + 1) Volume dependent	This involves the visual and physical quality control of incoming merchandise.	Damage pallet verification.	$(T_{Descharge} + T_{inspection}) \bullet X_i$
Fixed time (T_F) Volume independent	Administrative inspection process.	Signing off on the warehouse management system.	25

Table 7. Classification of Total Time Components into Fixed and Variable Activities.

$$\text{Service time } (T_{si}) = T_F + (T_{Descharge} + T_{inspection}) \bullet X_i$$

Where X_i represents the number of pallets of the specific store, which directly modifies the unloading and inspection time at each node, for example (using a high-capacity vehicle):

- If this truck only delivers 2 pallets, the unloading and inspection time will only take 5 minutes $2 \times (1.5 + 1)$.
- If that same truck goes to another node and unloads 20 pallets instead, the worker will take 50 minutes $20 \times (1.5 + 1)$.

Another limitation for the maximum capacity in terms of euro pallets is defined by the following inequality:

$$\sum_{Stores\ i \in Route_k} X_i \leq Q_k$$

This constraint implies that the sum of all pallets from the stores visited by the truck k cannot exceed the maximum capacity Q_k of its vehicle type *(indicated in Table 2*).

9. Daily Fixed Cost Parameter Matrix

To ensure the optimization model yields mathematically sound and logistically realistic topologies, the next table captures the activation fixed cost per vehicle type, according to depreciation per shift, mandatory insurance premiums, fixed labor blocks under strict German labor frameworks.

Vehicle type	Activation fixed cost	Justification
Small (S)	€ 180	Lower base cost. No heavy-vehicle tolls; standard-licensed driver (Class C1).
Medium (M)	€ 280	Increase due to higher asset depreciation, professional driver wages, and regional tolls.
Large (L)	€ 420	High cost due to strict axle restrictions, maximum highway tolls (Maut), and high usage penalties.

Table 8. Fleet Vehicle Profiles and Activation Cost Justification.

10. German Labor Compliance & Regulatory Frameworks (FPersV & EC No 561/2006).

According to the *Bundesamt für Logistik und Mobilität (BALM). (2025)*, under German social legislation for driving personnel (Fahrpersonalverordnung - FPersV) and European Union Regulation (EC) No 561/2006, labor costs in logistics cannot be modeled purely as a linear function of driving hours. Drivers must be compensated based on guaranteed shifts, creating a massive fixed financial block.

- Licensing and Qualification Overhead: Operating a Large (L) vehicle (a 40-tonne articulated tractor-trailer) requires a Class CE commercial license paired with continuous professional training under the Berufskraftfahrer-Qualifikation-Gesetz (BKrFQG). The scarcity of highly qualified CE drivers in the North Rhine-Westphalia cluster drives up the base reservation wage. Conversely, Small (S) vehicles (< 7.5 tonnes) can be operated by drivers holding simpler Class C1 or legacy Class 3 licenses, which command a significantly lower labor baseline.

- Temporal Hard Constraints: Because German law strictly mandates 45-minute rest breaks after 4.5 hours of driving and enforces rigid daily rest periods, a carrier must absorb the full cost of this highly regulated labor block the moment a vehicle leaves the depot.

Infrastructure Tolling Dynamics

The geographic positioning of Dr. Oetker's primary distribution networks in Bielefeld forces the routing model to navigate high-density federal corridors, primarily the A2 (East-West axis) and the A33 (North-South axis), along with key federal highways (Bundesstraßen like the B61 and B68).
The vehicle fixed activation costs ($FixedCost_m$) have been baseline-parameterized to reflect the daily operational amortizations, German labor standard allocations (Fahrpersonalverordnung), and the federal highway toll system (Lkw-Maut) typical of the Ostwestfalen-Lippe (OWL) region.

A high-capacity asset has a strict activation penalty of €420.00 to account for specialized commercial driver licensing requirements and maximum multi-axle road toll fees across the A2 and A33 autobahn corridors. Conversely, agile urban assets are parameterized at €180.00. This highly differentiated cost matrix ensures that Gurobi evaluates a realistic mathematical trade-off between fleet deployment penalties and spatial distribution efficiency under strict delivery time windows.

The Federal Tolling System (Toll Collect): Germany's federal road toll (Lkw-Maut) scales based on weight, axle configurations, and CO_2 emission classes.

- Small (S): vehicles under 3.5 tonnes are entirely exempt from the Maut, and those up to 7.5 tonnes incur minimal or zero charges on local networks, making their fixed activation baseline highly economical for urban or short-range drop-offs.
- Large (L): (typically 5 axles, 40 tonnes) are heavily penalized. Following recent legislative reforms, German toll rates include hefty carbon-pricing surcharges per kilometer.

Besides, by establishing $FixedCost_L$ at €420, the model introduces an infrastructural barrier that prevents the solver from routing a heavy truck across the A2 corridor unless the load consolidation heavily justifies the mandatory tolling overhead. In this way, this structure penalizes capacity redundancy, forcing Gurobi to achieve optimal Vehicle Fill Rates and ensuring that large assets are only activated when spatial-temporal consolidation reaches maximum efficiency.

IV. Mathematical Model

Decision variables

- Type of vehicle (low-, medium-, or high-capacity) according to the delivery route.
- The sequence in which the Lidl stores should be visited.
- Consequently, the number of vehicles and drivers required for the distribution process.

Sets and Indices

Set	Index	Description
N	i,j	Total system nodes. Including all the existing nodes in the area (1 distributor, Dr. Oetker, 24 Lidl stores): $N = \{0, 1, 2, 3 \ldots n\}$, where n = 25
S	i,j	This subset isolates the stores from the central depot (node 0) within the global node set N. This subset is essential because the depot operates under entirely different logical constraints than the commercial destinations. While stores exhibit specific material demands (Xi > 0). This prevents the solver from generating redundant loops and increases efficiency by eliminating unnecessary "if" conditionals. $S = N \setminus \{0\}$ or $S = \{1, 2, 3 \ldots n\}$,, where n = 24 This implies that the set excludes the origin node (Node 0).
K	k	A set of available vehicles inside the heterogeneous fleet, where $K = \{1, 2 \ldots K\}$
M	m	Type (Category) of vehicle $M = \{"Low", "Medium", "High"\}$
K_m	$m \in M$	A disjoint subset of individual vehicles belonging strictly to the capacity category mentioned before. $\bigcup_{m \in M} K_m = K \qquad K_m \cap K_{m'} = 0, \forall m \neq m'$ $\bigcup_{m \in M}$: Each and every truck in the fleet (K) belongs to one of the three capacity categories. There's no truck in the system considered left out of parameter restrictions $\cap$ = 0: An individual truck cannot belong to two categories at once. It is physically impossible for a vehicle to belong to multiple categories simultaneously.

Parameters

d_{ij}: Distance in kilometers (km) between node *i* and node *j*. It is assumed that d_{ii} *or* d_{jj} = 0.

t_{ij}: Time of travelling in minutes (min) between node *i* and node *j.* Calculated based on the average speed of the road network in the area (Google Maps data retrieved).
t_{ik}:Chronological arrival time (in minutes elapsed since the start of the shift) of vehicle k at node i.
α: (Buffer-Traffic) Road safety and congestion scaling coefficient in the OWL region.
X_i: Discrete supply demand for each Lidl store $i \in S$ measured in quantity of pallets (units). For the central depot evidently, is X_0: 0
v_{im}: Accessibility matrix (binary):

- 1 = If the vehicle m has physical access to the store $i \in S$
- 0 = If the vehicle is prohibited from accessing that store i.

Q_m: Maximum pallet capacity ($Q_{Low} = 15$, $Q_{Medium} = 24$, $Q_{High} = 33$) for type of vehicle.
$S_{max\,m}$: The maximum number of stores that the type of vehicle m can visit in a single route.

u_{ik}: It is a continuous variable representing the number of pallets remaining in truck k. immediately after leaving store i.

- 1 = If the vehicle m has physical access to the store i.
- 0 = If the vehicle is prohibited from accessing to that store i

For example, $v_{im} = 0$ would be when a high-capacity truck cannot have access to the historical center.

$D_{max\,m}$: Maximum route duration (405 minutes) per all categories.
c_m: Variable gas cost in euros (€/km) depending on type of truck and emission penalty measured in euros per kilometer traveled. $c_{Low} = 0.25$, $c_{Medium} = 0.38$, $c_{High} = 0.56$
T_f: Fixed time of service for the administrative procedure. $T_f = 25$ in minutes.
$t_{discharge}$: Variable time in minutes for pallet discharge. (1.5 min/pallet).
$t_{inspection}$: Variable time in minutes for pallet inspection. (1 min/pallet).

e_i: Lower bound of the time window (earliest arrival time) at store i.
l_i: Upper bound of the time window (latest arrival time) at store i.

Variable decision parameters:

- y_{ijk}: $\{0, 1\}$ Binary variable for the route.
    - 1 = If the vehicle k goes directly from node i to node j.
    - 0 = Otherwise

- u_{ik}: Continuous variable of the subroute. Indicates the order of visit sequence of the store i by the truck k. $u_{ik} \geq 0$ to prevent isolated loops.
- w_k: $\{0, 1\}$ Binary fleet usage variable.
  - 1 = If the vehicle k is activated to work.
  - 0 = If it is stored in the main deposit.

Objective Function

The goal of this mathematical function is to minimize the variable financial cost of distribution, resulting from the fuel consumption of the heterogeneous fleet over the distances traveled in direct proportion.

$$Min\, Z = \sum_{i \in N} \sum_{j \in N, j \neq i} \sum_{m \in M} \sum_{k \in K_m} c_m \bullet d_{ij} \bullet y_{ijk}$$

Breakdown

1. $Min\, Z$: Optimization criterion is the minimization, which represents the economic and environmental cost of daily operation.

2. $\sum_{i \in N} \sum_{j \in N, j \neq i}$: Double summation over the network's adjacency matrix that checks every possible trip from origin node i to a destination node j. The condition $j \neq i$ ensures that a node cannot be treated as its own destination, preventing optimization from creating self-loops or assigning travel costs from a node to itself. Within the summation, the multiplication by the adjacency matrix acts as a filter: only origin–destination pairs connected by an existing network link contribute to the objective function, while non-connected pairs contribute zero. The double summation then aggregates the contribution of all valid origin–destination pairs across the network.

3. $\sum_{m \in M} \sum_{k \in K_m}$ **:** Double summation for the heterogeneous fleet. It checks that each category of vehicle m (Low, Medium, High), and sequentially, each individual physical vehicle k belongs to the subset K_m
4. c_m : Weighted Cost Coefficient, which not only represents the financial cost of fuel per kilometer.
5. d_{ij}: Physical distances in kilometers between nodes in the OWL region.
6. y_{ijk}: A routing variable that acts as the "switch" in the equation. If the truck k of type m travels within i and j, $y_{ijk} = 1$, the cost of that segment is added to the total ($c_m \bullet d_{ij} \bullet 1$). If the truck does not travel that distance, $y_{ijk} = 0$, nullifying the product and preventing any cost from being added to the system.

System Constraints

1. $\sum_{j \in S} y_{0jk} = w_k \; \forall m \in M, \forall k \in K_m$

2. $\sum_{i \in S} y_{0ik} = w_k \; \forall m \in M, \forall k \in K_m$

3. $\sum_{i \in N, i \neq 1} \sum_{k \in K} y_{ijk} = 1 \; \forall j \in S$

4. $\sum_{j \in N, j \neq 1} y_{ijk} - \sum_{h \in N, h \neq j} y_{ijk} = 0 \; \forall j \in S, \; \forall k \in K$

5. $\sum_{j \in S} \sum_{i \in N,\, i \neq j} X_j \bullet y_{ijk} \leq Q_m \bullet w_k \; \forall m \in M, \; \forall k \in K_m$

6. $\sum_{j \in S} \sum_{i \in N,\, i \neq j} y_{ijk} \leq S_m^{max} \bullet w_k \; \forall m \in M, \; \forall k \in K_m$

7. $\sum_{i \in N} \sum_{j \in N, j \neq i} (\alpha \bullet t_{ij}) \bullet y_{ijk} + \sum_{j \in S} (T_f + (t_{discharge} + t_{inspection}) \bullet X_j) \bullet (\sum_{i \in N,\, i \neq j} y_{ijk}) \leq D_m^{max} \bullet w_k$

$\forall m \in M, \forall k \in K_m$

8. $u_{ik} - u_{jk} + Q_m \cdot y_{ijk} \leq Q_m - X_j + (Q_m - X_i - X_j) \cdot (1 - y_{ijk})$

$\forall i \in N, \forall j \in N, \forall j \in S, i \neq j, \forall m \in M, \forall k$

9. $0 \leq u_{ik} \leq Q_m \bullet w_k \; \forall i \in N, \; \forall m \in M, \; \forall k \in K_m$

10. $t_{ik} + T_f + (t_{\text{discharge}} + t_{\text{inspection}}) \cdot X_i + \mu(t_{ik}) \cdot t_{ij} - t_{jk} \leq M \cdot (1 - y_{ijk})$

$\forall i \in N, \forall j \in S, \forall j \in S, \; \forall k \in K_m$

11. $e_i \cdot w_k \leq t_{ik} \leq l_i \cdot w_k \quad \forall i \in S, \forall m \in M, \forall k \in K_m$

Variables:

$$y_{ijk} \in \{0,1\} \quad \forall i, j \in N, \forall k \in K$$

$$w_k \in \{0,1\} \quad \forall k \in K$$

$$u_{ik} \geq 0 \quad \forall i \in N_0, \forall k \in K$$

It constrains the algorithm's search space to prevent absurd solutions from the solver, such as using "0.5 vehicles or -1 trips" between supermarkets.
Therefore, routing and fleet usage decisions are binary, whereas the sequential sub-route counters are positive continuous variables.

Indexes

$i \in N$:Index representing the origin node of a route (which may be the distribution center (DC) or a Lidl store).
$j \in N$:Index representing the intermediate destination node or arrival stop visited by a vehicle.
$h \in N$: Secondary index representing the subsequent destination node (departure node) to which a vehicle travels immediately after completing a delivery.

Explanation of System Constraints

R1: Mandatory exit from the Central Depository

The origin node is fixed in i = 0 (Dr. Oetker Central Depository). The summation evaluates all possible destinations of the set of store, and the same equation is repeated for each vehicle k inside the heterogeneous fleet K_m.

- If vehicle k is activated (w_k=1) , consequently $\sum_{j \in S} y_{0jk}$ =1.
- If the vehicle is not used (w_k= 0), the restriction means that the truck cannot enter or exit from the deposit.

R2. Mandatory Return to the Central Depository

The destination node is fixed in j = 0 (Dr. Oetker Central Depository). The summation evaluates all possible origin nodes of the set of stores from which the vehicle can return.

- The truck is required to close its cycle by generating one arc leading back to the central depot. This ensures that no route remains open once deliveries are complete.

R3: Total Coverage and Single Customer Visit

By setting this constraint strictly equal to 1, the solution space is restricted so that each store is visited exactly once by a single truck. If an inequality (≥1) were used instead, the model would allow two different trucks to visit the same store to split the order, which would violate Lidl's logistics policy of avoiding duplicate loading dock costs and administrative paperwork.

R4: Flow Conservation at Commercial Nodes

It is evaluated for every combination of store j ∈ S and vehicle k ∈ K. The index i represents the nodes entering j, while the index h represents the nodes departing from j.

- **Active Transit (1−1= 0)** : If vehicle k is assigned to serve store j from an origin node i, the incoming binary variable is activated ($y_{ijk} = 1$). To satisfy the equality to zero, the solver is then forced to activate exactly one outgoing binary variable ($y_{jhk} = 1$). This ensures route continuity by requiring the vehicle to proceed to a subsequent destination node h.
- **Inactive (0 - 0 = 0):** If the optimal route of vehicle k does not include store j, all binary variables associated with that node take the value zero. The flow balance is therefore satisfied by default, allowing the vehicle to travel through other regions of the network without affecting the connectivity matrix of the unvisited nodes.

R5: Fleet Maximum Load Capacity Constraint

This constraint prevents the physical overloading of a vehicle.

- If truck k visits store j ($y_{ijk} = 1$), the store's pallet demand is accounted for X_j • 1. The summation, therefore, accumulates the total pallet demand of all stores assigned to the route of that specific truck.
- If the truck is inactive ($w_k = 0$), the right-hand side becomes 0, ensuring that no load is assigned to it. If the truck is active ($w_k = 1$), the upper bound is strictly set to the nominal capacity of its corresponding vehicle category $Q_m = (15, 24, 33\ pallets)$.

R6: Commercial Stop Limit per Route Constraint

This constraint counts a value of "1" each time an arc leading to a commercial store is activated. The sum of all inbound commercial arcs, therefore, represents the exact number of delivery stops assigned to a route.

- By bounding this quantity with $S_m^{max} \bullet w_k$, the solver is prevented from generating excessively long routes that could lead to driver fatigue or compromise product freshness. The maximum allowable number of stops is adjusted according to the corresponding truck category m.

R7: Traffic-Adjusted Workday Time Window

- If truck k visits store j, the corresponding assignment variable equals 1, activating the service time at that location. The service time consists of a fixed administrative component plus a variable unloading and inspection time that depends on the number of pallets delivered to the store. This service time is added to the total driving time of the route.
- To account for traffic congestion in the OWL region, all travel times are multiplied by a safety factor of α (depending on the traffic congestion factor for each node). The resulting total route duration (including both driving and service times) must not exceed the maximum allowable working time for the driver. This upper bound is enforced only when the truck is active (wk=1); otherwise, the right-hand side becomes zero, ensuring that an inactive vehicle cannot be assigned a route.

R8: Load Propagation Constraint

The continuous variable tracks the amount of cargo remaining on a truck as it moves along its route.

- If a connection between two locations is not part of the selected route, this constraint has no practical effect and does not influence the solution.

However, when the truck actually travels from one location to the next, the constraint ensures that the remaining load is updated correctly. After serving a store, the truck must carry fewer pallets than before, with the reduction matching exactly the number of pallets delivered at that stop.
As a result, the truck's load decreases progressively along the route, guaranteeing that deliveries are made consistently and that the remaining cargo is always physically feasible. This prevents the model from creating unrealistic routes in which the truck's load does not change after completing deliveries.

R9: Residual Load Boundary Conditions

This constraint defines the valid range of the residual load carried by a truck at any point along its route. The remaining load can never be negative, since a truck cannot deliver more pallets than it carries. Likewise, it can never exceed the truck's maximum loading capacity.

- If a truck is not used, the constraint forces its residual load to remain zero throughout the model. Consequently, only active trucks are allowed to carry cargo, while inactive trucks are assigned no load.

R10: Dynamic or Time-Dependent Vehicular Congestion Factor:

If a vehicle departs during critical morning commuter rush hours or late-afternoon industrial shift changes, the system automatically penalizes the transit duration by applying a higher time-dependent multiplier. Conversely, if the trip occurs during off-peak hours, travel time is computed under fluid baseline conditions. This prevents the optimization solver from generating theoretical routing strategies that are impossible to execute in practice due to real-world gridlocks.

R11: Hard Time Windows

This constraint was introduced to rigorously force the solver to respect these operational scheduling boundaries. By embedding a continuous chronological tracking clock variable for each vehicle, the model guarantees that if a truck arrives prior to the official opening of a store's time window, it logically incurs a waiting time parameter, while strictly prohibiting deliveries after the permitted closing threshold. Furthermore, the model is programmed to intelligently deactivate chronological progression for vehicles that remain idle at the central depot during the shift, thereby preserving an absolute mathematical consistency across the entire itinerary auditing matrix.

Mathematical Model Justification through Literature Review

To ensure the mathematical model developed for the Ostwestfalen-Lippe (OWL) logistical network remains mathematically consistent, constraints have been derived and justified utilizing key literature frameworks.

Depot Management and Structural Matrix Continuity:

- R1 (Mandatory Exit from Central Depository)
- R2 (Mandatory Return to Central Depository)
- R4 (Flow Conservation at Commercial Nodes)

The structural integrity of any routing graph relies on the rigorous mapping of origin-destination sequences. According to Toth and Vigo (2014) in Vehicle Routing: Problems, Methods, and Applications, a multi-index arc-based formulation utilizing a binary decision variable (y_{ijk}) requires strict mathematical boundaries to ensure complete network graph consistency.

R1 & R2: utilize the structural convention established by (*Irnich et al., 2014)*, fixing the graph coordinates at i = 0 and j = 0 as the single source and sink node (the Dr. Oetker depot). This mathematical bounding ensures that every active vehicle ($w_k = 1$) initiates a single independent subtour matrix and closes its cycle, preventing open trajectories or stranded vehicle anomalies.

R4: operationalizes structural flow conservation through an explicit three-index mathematical balance As validated by Irnich et al. (2014), the introduction of a third destination index (h) prevents "reflexive loop anomalies" by forcing the optimization solver to compute exact linear entry-exit pairings for every vehicle k at every commercial node $j \in S$.

Solution Space Restrictiveness and Policy Adherence

- R3: (Total Coverage and Single Customer Visit).

Traditional Capacitated VRP models occasionally relax customer visits using a loose inequality (≥ 1). However, to ensure operational feasibility within highly dense grocery supply chains, this model enforces a strict partitioning equality (= 1).

According to *Holzapfel, Hübner, and Kuhn (2016)* in Delivery Operations in Grocery Retailing manage tight logistics windows where multiple deliveries at a single docking bay create administrative redundancies, duplicate labor costs, and operational friction. Restricting the solution space to exactly equal 1 aligns the mathematical decision boundaries with the organizational policy requirements analyzed in German retail environments.

Heterogeneous Fleet Allocation and Operational Capacities

- R5: Fleet Maximum Load Capacity Constraint
- R6: Commercial Stop Limit per Route
- R9: Residual Load Boundary Conditions

The inclusion of non-homogeneous vehicular assets transforms a standard routing framework into a complex Rich VRP.

- R5 and R9 incorporate vehicle-specific capacity bounds (*Koç et al. 2016) and (Baldacci, Battarra, and Vigo 2008)* demonstrate that when fleet capacities vary significantly, capacity boundaries must be coupled with binary vehicle activation variables (w_k). This coupling ensures that the Gurobi optimization solver evaluates the economic trade-off between the high fixed costs of activating heavy assets versus the spatial efficiency of consolidating routes.
- R6 imposes a commercial stop limit dynamically adjusted by vehicle classification. This constraint is strongly supported by the ergonomics and commercial distribution research of *(Cáceres-Cruz et al., 2014)*, who state that realistic rich VRP formulations must prevent the optimization from overloading a single driver's shift, thereby acting as a practical safeguard against driving fatigue and securing food safety standards.

Chronological Progression and Big-M Tightening

- R8: Load Propagation Constraint

To prevent the mathematical generation of isolated routing components separate from the depot (known as the "independent subtour anomaly")the model must explicitly track cargo progression.

Our formulation relies on the continuous variable representing the remaining pallets immediately after leaving store i. This logic follows the classic *Miller-Tucker-Zemlin (MTZ)* subtour elimination relaxation framework. However, basic MT

To resolve this, we implemented the *(Desrochers and Laporte 1991)* Valid Inequalities to tighten the Big-M parameters:

$$u_{ik} - u_{jk} + Q_m \cdot y_{ijk} \leq Q_m - X_j + (Q_m - X_i - X_j) \cdot (1 - y_{ijk})$$

As justified by *(Toth and Vigo 2014)*, this advanced algebraic refinement significantly restricts the linear programming (LP) fractional convergence space. By mathematically "tightening" the Big-M relaxation weakness, Gurobi spends significantly less time processing deep Branch-and-Bound trees, preventing the optimization curve from stalling during large-scale stress testing instances.

Time-Dependent Velocity and Dynamic Regional Congestion

- R7: Traffic-Adjusted Workday Time Window
- R10: Time-Dependent Vehicular Congestion Factor

Standard VRP frameworks assume static travel times, which introduces massive scheduling risks in real-world distribution networks. To capture the logistical realities of the

Ostwestfalen-Lippe road network, travel times are operationalized using a time-dependent step-function multiplier

This dynamic temporal rule is mathematically grounded in the seminal work of *Ichoua, Gendreau, and Potvin (2003)* (Vehicle Routing with Time-Dependent Travel Times), who proved that time-slot variations must be scaled based on the exact epoch a vehicle departs from its origin vertex. Furthermore, the maximum workday limit acts as a hard operational barrier. Within Germany and the broader European Union, commercial freight transit must strictly comply with labor frameworks governed by the Fahrpersonalverordnung (FPersV) and EC Regulation No 561/2006. As documented by *Goel (2009)*, embedding these legal limits directly into the travel time calculations converts a simple routing matrix into a highly constrained, operational scheduling environment.

Hard Time Windows and Temporal Clock Tracking

- R11: Hard Time Windows

Retail grocery operations require highly precise delivery scheduling. Our model enforces strict temporal windows utilizing continuous chronological tracking variables (t_{ik}).

The implementation of hard time windows is supported by the authoritative taxonomies of *Solomon (1987) and Bräysy and Gendreau (2005)*. According to their research, if a vehicle arrives before the opening threshold , it must legally incur a waiting time penalty parameter, whereas any arrival post the closing boundary triggers an automatic mathematical infeasibility. By hard-coding these boundaries into the solution space, the solver acts as an absolute strategic decision-support filter, ensuring that computed itineraries match the rigid, real-world Time-Slot Management Software configurations utilized across modern Lidl distribution centers.

Solver Approach Type : Mixed-Integer Linear Programming (MILP)

The model is classified as mixed because it combines binary decision variables, which determine whether a truck travels between two locations, with continuous variables, which represent the amount of cargo remaining on each truck throughout its route (Fachrizal, 2020). The problem is computationally complex because it simultaneously involves deciding how to assign customer demands to vehicles while determining the optimal sequence of deliveries. These decisions combine characteristics of the *Bin Packing Problem* and the *Traveling Salesman Problem*.

Trade-offs

The optimal solution is not straightforward because several real-world factors create conflicting objectives and limitations within the routing and fleet assignment problem:

- High-capacity vehicles may reduce the number of drivers required but could increase unloading times or reduce route flexibility.
- Smaller vehicles may improve maneuverability and scheduling flexibility, although they require more drivers and trips.
- The shortest geographical distance between two points is not always the fastest route, since real road networks are not linear and include turns, intersections, and indirect paths.
- Routes that appear shorter may pass through highly congested roads, leading to longer travel times compared to longer but faster alternative routes.
- Certain streets may be too narrow or restricted for specific vehicle types, making some theoretically shortest routes infeasible.
- Road infrastructure constraints such as toll roads, traffic signals, pedestrian zones, or delivery restrictions can affect route feasibility and cost.
- Delivery time windows can force vehicles to take longer detours in order to arrive at stores at a feasible time.
- Loading and unloading constraints, including ergonomic llimitations, may increase service time at certain stops and influence route sequencing decisions.
- Traffic variability during the day means that the same route can have different travel times depending on departure time.

Because of these interacting factors, the "optimal" route is not simply the shortest or low-cost one, yet the one that achieves the best balance between time, cost, feasibility, and operational constraints.

Limitations

This section evaluates the primary missing factors, analyzes how their inclusion could alter the strategic recommendations, and outlines the mathematical frameworks required to incorporate them into future model extensions.

1. **Deterministic Travel Times vs. Real-Time Traffic Stochasticity**

The current model treats travel times between nodes as deterministic parameters, utilizing historical averages scaled by a discrete, piecewise time-dependent step-function multiplier to simulate a traffic congestion in the OWL region (e.g., along the A2/A33 autobahns and bottlenecks like Herford Straße). Although, in reality, transportation networks are subject to severe stochasticity caused by non-recurrent congestion events, such as traffic accidents, sudden weather anomalies, or highway construction delays.

1.1 Impact on Recommendations

By omitting real-time stochasticity, the model assumes a <u>level of predictability that does not exist in daily distribution</u>. For example, if a high-capacity truck (Type-High) experiences an 30-minute delay due to a crash at the Kreuz Bielefeld, it will systematically violate the rigid, hard time windows of down-route Lidl stores.

This triggers operational infeasibility in practice, leading to missed delivery slots, financial penalties from retail managers, and cold-chain compliance failures. If stochasticity were included, the recommendation would likely pivot away from tightly packed, maximum-capacity routes toward more conservative, decentralized routing sequences using smaller, agile vehicles (Type-Small) that absorb delays more easily.

1.2 Methodological Extension
To integrate this factor, the deterministic formulation must transition into a *Stochastic Vehicle Routing Problem with Time Windows (SVRPTW*). Travel times would be modeled as random variables following a continuous probability distribution (e.g., log-normal or Gamma). Mathematically, the hard temporal constraint tracking arrival time would be replaced with a *Chance-Constraint Programming (CCP) formulation:*

$$\mathbb{P}\left(t_{ik} \leq l_i\right) \geq \alpha \quad \forall i \in S, \forall k \in K_m$$

2. **Dynamic Order Fluctuations**

The demand data for each Lidl store is assumed to be deterministic and fully known prior to the optimization run. This assumption is justified by the widespread use of Electronic Data Interchange (EDI) systems in German grocery retailing *(Holzapfel et al., 2016)*. However, this approach omits critical operational dynamics, such as same-day supply chain fluctuations, sudden intra-day stockout corrections, and dynamic order cancellations that occur between the tactical planning phase and actual truck dispatching. Furthermore, it fails to account for seasonal variations in consumer behavior.

2.1 Impact on Recommendations
If a store's demand unexpectedly surges beyond its deterministic EDI forecast while a truck is mid-route, the assigned vehicle may experience a capacity violation. This forces the driver to return prematurely to the Dr. Oetker depot in Bielefeld to reload, <u>completely destroying the pre-calculated spatial alignment and doubling variable fuel costs.</u> Conversely, if demand is lower than expected, the model over-allocates high-capacity assets (Type-High= €420 in daily activation) when cheaper, low-capacity urban assets (Type-Small = €180) would have sufficed, leading to sub-optimal fleet utilization and unnecessary fuel costs.

2.2 Methodological Extension
This would be fixed if the model turns into a *Vehicle Routing Problem with Stochastic Demands (VRPSD)*. Under this paradigm, demands are characterized by a known probability distribution. The flow and load balance constraints would be adjusted to account for the risk of a route "failure" (when the cumulative demand on a route exceeds the pallet capacity). The objective function would be enhanced to minimize the expected cost of recourse, which mathematically represents the expected distance of traveling back to the depot mid-route.

3. **Multi-Period Rolling Horizon Allocation**

3.1 Missing Factors

The current planning horizon is strictly bound to a single operational day (a discrete daily rolling horizon). It assumes that at the end of the shift, all activated vehicles return to the central Dr. Oetker depository, reset their status, and the drivers complete their shifts within the maximum allowed duty duration. It does not model asset re-allocation, weekly driver shift rotation, or multi-day inventory levels at the retail destinations.

3.2 Methodological Extension

To address this, the decision domain must scale into a Multi-Period Vehicle Routing Problem (MPVRP) or an Inventory Routing Problem (IRP). This requires adding a discrete time-horizon index $t \in T$ (where $T = \{1, 2, \dots 7\}$ days) to all variables, parameters, and constraints. A continuous inventory tracking variable would be introduced for each store.

Future Extensions

CO_2 Emissions Generated by Vehicle Type

To support a more sustainable approach, future iterations of this model can incorporate additional decision variables that align with Industry 5.0 principles. While these variables were excluded from the current scope due to time constraints and the risk of excessive computational complexity, they represent a vital next step. Moving forward, the following mathematical structures will be developed using the datasets detailed below to transition the model from a strict cost-minimization focus to a holistic framework that mitigates environmental impact.

To calculate the environmental cost or constraint bounds within a *Time-Dependent Vehicle Routing Problem (TDVRP),* carbon dioxide emissions are treated as a direct function of travel distance, scaled by vehicle-specific fuel consumption coefficients under regional infrastructure stress (Toth & Vigo, 2014). Following standard European carbon auditing frameworks (COPERT / European Environment Agency), burning one liter of diesel fuel releases a fixed constant of 2,640 grams of CO_2.

The heterogeneous fleet allocated for the OWL-Lidl delivery network is broken down below with its respective empirical consumption rates and emission factors:

Fleet Class	Fuel Consumption (L/100 km)	CO_2 Emissions Factor
Small (S)	~11.5 L / 100 km	0.304 kg CO_2 / km
Medium (M)	~22.0 L / 100 km	0.581 kg CO_2 / km
Large (L)	~34.0 L / 100 km	0.898 kg CO_2 / km

Table 9. Breakdown of Fuel Consumptions and CO_2 / km per each fleet class.

Introducing as a new index:

E^{max}: CO_2 emission coefficient per kilometer traveled for vehicle type m.

and a new constraint like the following:

12. $$\sum_{m \in M} \sum_{k \in K_m} \sum_{i \in N} \sum_{j \in N, j \neq i} \epsilon_m \cdot d_{ij} \cdot y_{ijk} \leq \mathcal{E}^{\max}$$

Without a mathematical boundary acting as a direct environmental throttle, the optimization system will invariably converge toward the cheapest routing permutations, even if that entails saturating the regional ecosystem with a highly elevated carbon footprint.

This new constraint functions as an absolute daily carbon budget for the distribution network. By mapping specific emission coefficients per kilometer traveled based on each truck's capacity and engine classification, the model quantifies the cumulative fleet emissions. If a proposed deployment schedule exceeds the maximum allowable pollution threshold, the entire routing plan is deemed infeasible. This forces the algorithm to act as a sustainability filter, compelling it to maximize the spatial clustering efficiency of store sequences and prioritize the allocation of eco-efficient assets to avoid violating the environmental cap imposed by the planning horizon requirements.

Accessibility physical restriction

Another critical element for future development is the incorporation of physical accessibility restrictions. In the OWL area, large vehicles exceeding 35 tonnes are prohibited on certain streets. Integrating these routing constraints into the current model would significantly elevate its computational complexity, placing it beyond the current project's scope. However, this extension should be developed in a future phase. The proposed mathematical constraint, along with the logic for classifying access restrictions per retail store, is structured as follows:

13. $$\sum_{j \in N, j \neq 1} y_{ijk} \leq v_{jm} \forall j \in S, \forall m \in M,\ \forall k \in K_m$$

- If store j (for example, in an OWL-restricted historic city center) prohibits high-capacity trucks, the parameter is set to $v_{j\text{"High"}} = 0.$ The inequality then becomes ≤ 0, which immediately eliminates any inbound arc to that store served by a heavy vehicle.
- If the store has permitted access $v_{jm} = 0.$, the inequality then becomes ≤ 1, making the solver freely decide if that route can be used or not.

The permitted fleet for each node is divided according to a technical justification based on the node's geographical location. In the red-coded zones, only low-capacity vehicles are permitted. Meanwhile, the orange-coded zones offer higher flexibility, allowing both low and medium-capacity vehicles to access the nodes.

Node	Location	Road Zone Type (Industrial, Open, Historic, Residential)	Type of fleet permitted	Technical Justification
N_0	Lutterstraße 14 · 33617 Bielefeld	Industrial zone	All categories	Designed for standard heavy-duty logistics.
N_1	Teutoburger Straße 98, 33607 Bielefeld-Mitte	Historic / Central urban	Low capacity only	Located in a dense urban center with strict axle weight and geometric turning radius constraints. High/Medium trucks are legally/physically restricted.
N_2	Jöllenbecker Straße 41, 33613 Bielefeld-Mitte	Historic / Central urban	Low capacity only	High-density urban main street with restricted loading bays and narrow approach corridors. High-capacity vehicles prohibited.
N_3	Heeper Straße 113, 33607 Bielefeld-Mitte	Residential zone	Low, Medium	Residential/commercial transition zone. Structural noise regulations and traffic calming measures prevent the use of high-capacity heavy-duty trucks
N_4	Stadtheider Straße 11, 33609 Bielefeld-Mitte	Industrial zone	All categories	Industrialized commercial area with dedicated wide access roads and high-tonnage docking bays.
N_5	Detmolder Straße 345, 33605 Bielefeld-Stieghorst	Open / Arterial road	All categories	Situated on a major arterial transit road with seamless highway connection. Fully accessible for all vehicle dimensions and capacities.
N_6	Babenhauser Straße 18, 33613 Bielefeld-Schildesche	Residential zone	Low, Medium	Dense residential periphery. High-capacity multi-axle trucks are banned due to low clearance, tight inner-district curves, and local noise protection ordinances.
N_7	Kimbernstraße 1, 33647 Bielefeld-Brackwede	Open road	All categories	Outskirts commercial zone. Broad approach roads allow unrestricted maneuvering for high-capacity trucks without urban bottlenecks.
N_8	Oldentruper Straße 255, 33719 Bielefeld-Heepen	Industrial zone	All categories	Standard industrial park setup. Infrastructure built for heavy freight, including large turning loops and automated docking ramps.
N_9	Detmolder Straße 550, 33699 Bielefeld-Stieghorst	Open / Arterial road	All categories	Direct proximity to the federal highway interface. Exceptional structural accessibility with zero geometric or legislative barriers for the entire fleet.
N_{10}	Kasseler Straße 1,	Open road	All categories	Regional connecting route with wide lane layouts.

	33649 Bielefeld-Brackwede			Supports maximum gross vehicle weight, enabling full high-capacity truck operations.
N_{11}	Henleinstraße 1, 33689 Bielefeld	Industrial zone	All categories	Heavy-duty commercial grid. Optimized for heavy distribution logistics; no structural scaling down required for high-capacity vehicles.
N_{12}	Neuenkirchener Str. 100, 33332 Gütersloh	Open road	All categories	Major link road in Gütersloh. Generous spatial clearance and high load-bearing capacity for full-scale logistics fleets.
N_{13}	Carl-Bertelsmann-Straße 152, 33332 Gütersloh	Industrial zone	All categories	Proximity to massive corporate industrial complexes. Roadways engineered specifically for high-frequency heavy-tonnage vehicle flow.
N_{14}	Auf dem Stempel 10, 33334 Gütersloh	Residential zone	Low, Medium	Suburban residential hub. Subject to municipal weight caps and street-side parking narrowing, which makes high-capacity truck transit physically hazardous.
N_{15}	Westring 11, 33415 Verl	Open road	All categories	Outer bypass ring road. Free of historical urban cores or narrow street architectures, enabling uninhibited logistics for all vehicle sizes.
N_{16}	Bielefelder Str. 55, 33378 Rheda-Wiedenbrück	Historic/Central urban	Low capacity only	Enclosed historic urban layout with narrow medieval-derived road widths and strict local environmental/vibration laws. Only a low-capacity fleet is allowed
N_{17}	Berliner Ring 40, 33428 Harsewinkel	Open road	All categories	Major ring distribution route. Designed for cross-regional freight transit with expansive spatial lanes and high maneuverability indexes.
N_{18}	Lüchtenstraße 61, 33758 Schloß Holte-Stukenbrock	Residential zone	Low, Medium	Strict suburban residential zoning. Heavy environmental noise-mitigation caps enforce a strict ban on high-capacity, multi-axle freight vehicles
N_{19}	Elsa-Brändström-Straße 1, 33790 Halle (Westfalen)	Residential zone	Low, Medium	Strict suburban residential zoning. Heavy environmental noise-mitigation caps enforce a strict ban on high-capacity, multi-axle freight vehicles
N_{20}	Bahnhofstraße 23, 33803 Steinhagen	Historic / Central urban	Low capacity only	Core municipal retail street with tight parking clusters and restricted delivery windows. Physical geometries completely block high/medium-capacity units.
N_{21}	Krentruper Str. 34, 33818 Leopoldshöhe	Open road	All categories	Rural/open arterial infrastructure with expansive lanes. No geometric bottlenecks, allowing high-capacity trucks to minimize delivery runs.
N_{22}	Lange Str. 35-37, 32139 Spenge	Historic / Central urban	Low capacity only	Spenge's central corridor has narrow roadside lanes and structural overhangs. High-capacity vehicles are structurally excluded to prevent gridlock.
N_{23}	Friedrich-Petri-Straß	Residential zone	Low, Medium	Medium-density residential neighborhood. Low

	e 18-26, 32791 Lage			vertical clearance from trees/cables and tight street parking removes high-capacity vehicle options.
N_{24}	Ahmser Str. 50-54, 32052 Herford	Industrial zone	All categories	High-volume industrial bypass. Built to handle heavy freight traffic; fully compliant with high-capacity truck requirements
N_{25}	Lambernweg 46, 32130 Enger	Open Road	All categories	Unrestricted peripheral connecting road. Large layout allows optimal route consolidation using high-capacity vehicles without accessibility penalties.

Table 10. Accessibility physical restriction per each node, establishing road zone type, the type of fleet permitted, and the technical justification.

Appendices

App. 1
stochastic: "A stochastic process or system is connected with random probability" (Cambridge Dictionary, n.d).

App. 2. OWL area:

Image retrieved from: https://www.amazon.de/s?k=reisef%C3%BChrer+ostwestfalen+lippe&adgrpid=1206164911060879&hvadid=75385424524280&hvbmt=be&hvdev=c&hvlocint=116641&hvlocphy=129550&hvnetw=o&hvqmt=e&hvtargid=kwd-75385511156668%3Aloc-72&hydadcr=29569_2413232&mcid=2fea2a13ec653ea2a1b8eede075b618c&msclkid=a9fed498ddeb117b1bdbbeab726c5623&tag=hyddemsn-21&ref=pd_sl_956js848nk_e)

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